Energy 254 (2022) 124190

Contents lists available at ScienceDirect

Energy

journal homepage: www.elsevier.com/locate/energy

Comprehensive assessment of a green cogeneration system based on compressed air energy storage (CAES) and zeotropic mixtures

Hao Bai [a, b], ShiHao Luo [a], Xijie Zhao [c], Gen Zhao [b, *], Yang Gao [d]

[a] Sichuan Expressway Construction & Development Group Co.,LTD., Chengdu, 610041, China
[b] Shudao Investment Group Co.,Ltd., Chengdu, 610041, Chengdu, China
[c] Sichuan Egang Highway Engineering Construction Co., Ltd, China
[d] The University of Edinburgh,57 George Square, Edinburgh, EH8 9JU, UK

ARTICLE INFO

Article history:
Received 17 June 2021
Received in revised form
30 April 2022
Accepted 2 May 2022
Available online 14 May 2022

Keywords:
Compressed air energy storage
Ejector refrigeration cycle
Zeotropic mixtures
Exergoeconomic analysis
Payback period

ABSTRACT

This paper proposes combining the solar-based adiabatic compressed air energy storage system to respond to the grid peak-time demand. Two ejector refrigeration cycles with zeotropic mixture working fluid are employed as the waste heat recovery system. Eight different zeotropic mixtures were candidates for the first ejector refrigeration cycle working fluid. Its working fluid selection was carried out via studying their mass fraction variation on the system's performance. Among the considered zeotropic mixture, the Pentane/Butane with a mass fraction of 0.5–0.5 presented the best performance. Comprehensive thermodynamic and thermo-economic analyses were performed to evaluate the system's performance. The proposed plant could provide 991.89 kW power for the grid's peak-shaving and 170.7 kW cooling load with 35.82% and 33.19% round trip and exergetic round trip efficiencies and 7.75 years of payback period. A parametric study was performed to investigate some main parameters variation's effect on the system's performance indexes and demonstrated that the cavern inlet pressure has the highest impact on the system's power production, and charging to discharging pressure ratio influenced the round trip, exergetic round trip efficiencies. Finally, triple-objective optimization is applied, obtaining 35.87%, 33.23%, and 7.72 years of round trip efficiency, exergetic round trip efficiencies, and payback period.

© 2022 Elsevier Ltd. All rights reserved.

1. Introduction

Population growth, industrialization, and lifestyle change lead to increasing power demand. This demand varies in the daily timeframe, consisting of peak time. Responding to the peak time demand is an important challenge for all societies [1,2]. Some solutions have been proposed by researchers, in which the energy storage method attracts the attention considerably [3].

The energy storage concept is preferred for peak shaving, peak shifting, load leveling, energy management, and standby power in the form of spinning reserves [4]. These systems are employed to support on-grid power plants during real-time demand and are a promising alternative for fossil fuel-fed auxiliary systems for grid stabilization [5]. Compressed air energy storage (CAES), liquid air energy storage (LAES), thermal energy storage (TES), and pumped hydraulic storage (PHS) are well-known energy storage systems [6]. Due to its overt advantages like longevity, less environmental impact, reliability, availability, and economic benefits [7]. In the CAES system, the ambient air is compressed and conventionally stored in large underground caverns or artificial storage vessels (SV) during charging time, which is the power grid off-peak time [8]. Also, it can be utilized for various scales of power systems [9]. At the grid peak-demand time, the stored compressed is released from the SV and led to the turbine for generating power [10]. However, CAES systems include environmental issues such as CO_2 emission [11]. Adiabatic Compressed Air Energy Storage (A-CAES) has been proposed to eliminate this problem [12]. The main distinction between an A-CAES system and an ordinary CAES system is that a thermal energy storage (TES) system has been substituted with the combustion chamber and supplies the turbine inlet flow's desired temperature, which leads to zero environmental impact [13]. Another type of CAES is Isothermal-CAES. In the Isothermal-CAES system, the air is compressing due to multi-stage compression, and at each stage of compressing process, the air's heat is absorbed

* Corresponding author.
E-mail address: secdcbh@126.com (G. Zhao).

https://doi.org/10.1016/j.energy.2022.124190
0360-5442/© 2022 Elsevier Ltd. All rights reserved.

Nomenclature

A	Area [m^2]
A-CAES	Adiabatic Compressed Air Energy Storage
AS	Annual Saving [$]
C	Thermal capacity
c	Cost per exergy unit [$.$GJ^{-1}$]
CAES	Compressed Air Energy Storage
COP	Coefficient Of Performance
CRF	Cost Recovery Factor
D	Diameter [m]
DNI	Direct Normal Irradiation
e	Exergy [kJ.kg^{-1}]
$\dot{E}$	Exergy flow [kW]
ERC	Ejector Refrigeration Cycle
ERTE	Exegetic Round Trip Efficiency [%]
FES	Flywheel Energy Storage
Gen	Generator
h	Enthalpy [kJ.kg^{-1}]
HTES	High Temperature Energy Storage
IF	Inflation Factor
K	Interest rate
LAES	Liquid Air Energy Storage
LTES	Low Temperature Energy Storage
$\dot{m}$	Mass Flow Rate [kg.s^{-1}]
M	Mass [kg]
MM	Molar Mass [kg.$kmole^{-1}$]
MOPSO	Multi-Objective Particle Swarm Optimization
N	Number of collectors
NPV	Net Present Value [$]
Nu	Nusselt number
ORC	Organic Rankine Cycle
PHS	Pumped Hydraulic Storage
PTSC	Parabolic Trough solar collector
Q	Heat [kJ]
PP	Payback Period [years]
PR	Charging to Discharging Pressure Ratio
RDF	Real Discount Factor
RIR	Real Inflation Rate
RTE	Round Trip Efficiency [%]
s	Entropy [kJ.kg^{-1}.K^{-1}]
SV	Storage Vessel
t	Time [s]
T	Temperature [K]
TES	Thermal Energy Storage
V	Volume [m^3]
W	Work [kJ]
WHR	Waste Heat Recovery
Z	Investment cost of components [$]

Subscript and abbreviations

AC	Air Compressor
Ap	Aperture Area [m^2]
AT	Air Turbine
c	Collector
Ch	Charging
coll	Collector
Comp	Compressor
Cond	Condenser
COT	Cold Oil Tank
CWT	Cold Water Tank
DCh	Discharging
Elec	Electrical
Eje	Ejector
EV	Expansion Valve
Eva	Evaporator
HOT	Hot Oil Tank
HWT	Hot Water Tank
H.X	Heat Exchanger
in	Inlet
inst	Installation
invst	Investment
is	Isentropic
mix	Mixed
OM	Operation and Maintenance
out	Outlet
pf	Primary Flow
pp	Pintch Point
sf	Secondary Flow
TT	Terminal Temperature
Tur	Turbine
I	First
II	Second
0	Dead State
1,2,3, …	State Point

Greek symbols

ρ	Density
η	Efficiency
μ	Entertainment Ratio
ε	Effectiveness
γ	Intercept factor
τ	The transmittance of the glass cover
α	The absorbance of the receiver
K	Incidence angle modifier
φ	maintenance factor

by a cooling fluid and stored. At the discharging time, the absorbed heat is utilized to heat the compressed air before entering the air turbine [14]. The CAES systems can recover a significant amount of waste heat in compressors and turbines [15]. To recover the waste heat in compressors and turbines in CAES systems, different methods, improving the round-trip and exegetic round-trip efficiencies, have been suggested by researchers [16]. The organic Rankine cycle (ORC) is an all-purpose waste heat recovery (WHR) subsystem for CAES systems [17,18]. Hence, different studies have been performed to investigate the CAES systems' utility with the heat recovery systems.

1.1. Literature review

Hemmati et al. [19] investigated the utility of the CAES system to be responding to Khaf, Iran's real power demand, as a case study. Their results showed that employing CAES provided 11.36% more profit and reduced fossil fuel consumption by about 11.33%. Razmi et al. [20] studied a combination of biomass-based CAES systems. They investigated the performance of employing green waste, wood chips, and municipal solid waste as the biomass in their systems. Regarding their results, municipal solid waste represented the highest energetic efficiency. Mohammadi et al. [21] employed an organic Rankine cycle (ORC) with toluene as the working fluid together with an absorption refrigeration cycle to increase power

generation and produce the cooling capacity by recovering waste heat in a CAES system. Razmi et al. [22] proposed a CAES system to generate power for peak shaving. They added a compression-absorption refrigeration cycle to their CAES system and improved its RTE from 13.15% to 66.15%. Wang et al. [23] utilized an ORC cycle with R245fa working fluid to reduce air turbine temperature and increase the ORC inlet energy, leading to an improvement in the exegetic round-trip efficiency of the proposed system. Sadeghi and Ahmadi [24] presented a CAES system, which runs a supercritical CO_2-based CCHP system. Their system provided 136.56 MW power, 65.8 MW, and 1.96 MW heating and cooling capacities. Also, they applied a Multi-objective optimization, in which they achieved a 4817.7 $/h total cost rate and 68.19% exergetic efficiency. Alirahmi et al. [25] studied a combination of a solar-based Brayton cycle with a CAES system from thermodynamic and thermo-economic aspects, obtaining 60.4% ERTE with a total cost rate of 117.55 $/GJ at the optimum state. Javidmehr et al. [26] proposed a solar-based A-CAES system with an ORC and multi-effect desalination unit as a WHR unit at the micro-scale. Their system provided 532 kWh net power with 2.5 ton fresh water and achieved 65.2% and 70.35% round trip and exergetic round trip efficiencies at the optimum state. Zhang et al. [27] proposed a biomass-based CAES and employed an absorption chiller refrigeration cycle with a heat pump to recover the gas turbine exit flow's energy. The applied energy, exergy, and exergoeconomic analyses on their proposed system represented 90.06% round trip and 31.52% exergetic efficiency with 3.03 years payback period. Meng et al. [28] used a recuperator-equipped ORC as a subsystem in a CAES system. The authors compared the performance of five different organic working fluids and showed that R123 represents the best performance and increases the RTE by 6.7%. Razmi et al. [29] proposed a multi-effect desalination cycle combined with a CAES system. The results indicated that the proposed system produces 38 kg/s and 62.5 kg/s of freshwater during the grid off-peak and peak times, respectively, and also generates 80 MW of electricity to support the grid during the peak time. Sadreddini et al. [30] integrated an ejector refrigeration cycle (ERC) using toluene as the working fluid and ejector refrigeration with a CAES system and improved the RTE by around 3.87%. Razmi et al. [31] studied the combination of wind energy with the CAES system for peak-shaving of the grid demand. They considered Abhar and Kahak wind farms as the case study. They performed an energetic analysis to estimate their proposed system's performance, which led to obtaining an RTE of about 43%. Mousavi et al. [32] performed an energetic, exegetic, and exergoeconomic study of a cascade-packed bed thermal energy storage-based CAES system. They obtained 61.5% and 68.2% RTE and ERTE with 3.5 years payback period. Alirahmi et al. [33] studied a solar-based CAES system with a multi-effect desalination unit from exergoeconomic viewpoint. They considered the San Francisco city, the United States of America, as a case study. Their proposed system represented 226.87 m^3 freshwater with 27551 kWh electricity with the 2.67 years payback period and 112 M$ net profit during the system lifetime.

ORC and ERC subsystems involve a phase-change process in their heat exchanger unit; this process occurs at a constant temperature and imposes significant exergy destructions and temperature mismatches. To minimize the exergy destruction, zeotropic fluids are utilized as working fluids in these subsystems. Since the zeotropic fluids contain two organic fluids with different vaporization temperatures, the temperature does not remain constant during the phase-change process of the working fluid in the heat exchanger; therefore, the exergy destruction of the heat exchanger considerably decreases, and the temperature mismatch fades. Li et al. [34] compared the pure and zeotropic working fluids for an ORC. The outcomes revealed that the proposed ORC performance significantly improves by utilizing the zeotropic working fluids with an internal heat exchanger. Wang et al. [35] studied a solar Rankine cycle with different zeotropic working fluids and mass fractions. The authors deduced that the system's performance enhanced thanks to adding superheating and internal heat exchangers. Zhao et al. [36] drew an experimental comparison of the zeotropic and pure substance performance in an ORC. It was discovered that the zeotropic fluid of R245fa-R152a with a mass fraction of 0.7–0.3 exhibits a higher thermal performance compared to pure R245fa.

1.2. Main contribution and novelties

The importance of electricity storage is increasing as the awareness about the importance of more utilization of intermittent renewable energy increases. Energy storage technologies are mainly attractive due to their potential for load balancing in the electricity grid and storing the surplus power during peak production periods for later use at peak consumption times. Compressed air energy storage is one of the promising energy storage technologies being given much attention as it offers a very high energy density, a relatively low cost of capital, a short response time to load change, and competitive energy efficiency. CAES systems with manufactured air vessels are a more adaptable solution, especially for distributed generation that could be widely applicable to future power networks. In the case of the micro-CAES, it is possible to use the dissipated heat of compression for residential heating, which can contribute to improvements in energy efficiencies. In addition, compressed air systems can be used for both power generation and cooling load. This work proposes a new configuration of CAES technology aiming at cogeneration of electricity and cooling. The new system may be very advantageous for locations with high penetration of renewable energy in the electricity grid as well as high cooling demands. The latter would typically be locations with district heating and cooling networks.

Accordingly, an adiabatic CAES system is coupled with two ejector refrigerator cycle (ERC) subsystems to recover and convert the waste heat of the compression process and air turbine exit flow into cooling when operating at charging time. Comprehensive mass, energy exergy, and exergoeconomic analysis are performed to evaluate the proposed system performance via EES [37] and MATLAB [38] codes. Also, eight different zeotropic mixtures are selected for the ERCs subsystem's working fluid. Then a parametric study is performed to select the best zeotropic mixture for the ERC subsystem. Moreover, the following novelties have been considered in the current study:

- The zeotropic mixture selection through the parametric study illustrates a brief presentation of the ERC subsystem performance under various operational conditions.
- The zeotropic mixture's mass fraction is considered as the variable in the parametric study of the zeotropic mixture selection, and it leads to obtaining the best mass fraction of the mixture.
- A PTSC unit is employed to supply the air turbine desired inlet temperature.
- The net present value is estimated for three different incoming and products' sale prices.
- A triple-objective optimization is applied to the proposed system to evaluate its performance at the optimum state.

1.3. Paper structure

The present study contains different sections in which different conducted studies are placed at each section and classified into the following category:

Section 2 presents the system description in which the flow direction and conducted process in the proposed system are presented. Then, the mathematical modeling of the proposed system and employed concepts to simulate the system are illustrated in section 3. Then, the simulation's results are illustrated in section 4. In section 5, a triple-objective optimization based on the multi-objective particle swarm optimization (MOPSO) method is performed to optimize the proposed system's payback period, round trip, and exergetic round trip efficiencies. Finally, the overall conclusions of the current study are demonstrated in section 6. Also, the flowchart diagram of the present study is represented in Fig. 1.

2. System description

All CAES systems operate at charging and discharging times. The system starts to store the compressed air during the charging time, whereas during the discharging time, the stored compressed air is given to the turbine to generate extra power. The schematic diagram of the proposed system is shown in Fig. 2. As can be observed, the system consists of CAES and ERC subsystems. At charging time, the ambient air enters three-stage compressors, which are pressurized using the excess power from the grid. Along with increasing the pressure, the air temperature increases; such content of energy is transferred to the cooling water injected from the cold tank into the three coolers (state 24). Passing through the compressors, the ambient air is guided to the intercoolers, where it is cooled down by cooling water. The compressed air is finally stored in a cavern after getting pressurized in the third compressor (state 6) and cooled down in the aftercooler (state 7). The cooling water, which is now hot thanks to absorbing the compressed air's heat, is sent to the ERC vapor generator (state 30). The cooling water crosses the vapor generator and transfers a part of its energy to the ERC working fluid, and then it is stored in the hot tank (state 31). In the ERC 1 subsystem, the zeotropic working fluid is vaporized by receiving energy in the vapor generator (state 13) and then injected into the ejector and entrained evaporator outlet flow. The ejector's structure and its h-s diagram are shown in Fig. 3a and b, correspondingly. The ejector outlet flow refers to the condenser being cooled down (state 14). Leaving the condenser, the ERC working fluid is divided into two parts; the first part is sent to pump 1 to reach the vapor generator pressure (state 16) and receive the cooling water energy, while the second part passes through the throttling valve (state 17). The T-s diagram of the ERC 1 subsystem is illustrated in Fig. 3c.

After passing the throttling valve, the ERC working fluid goes into the evaporator, generates the cooling capacity (state 18), and leaves the evaporator to the ejector (state 19). On the other hand,

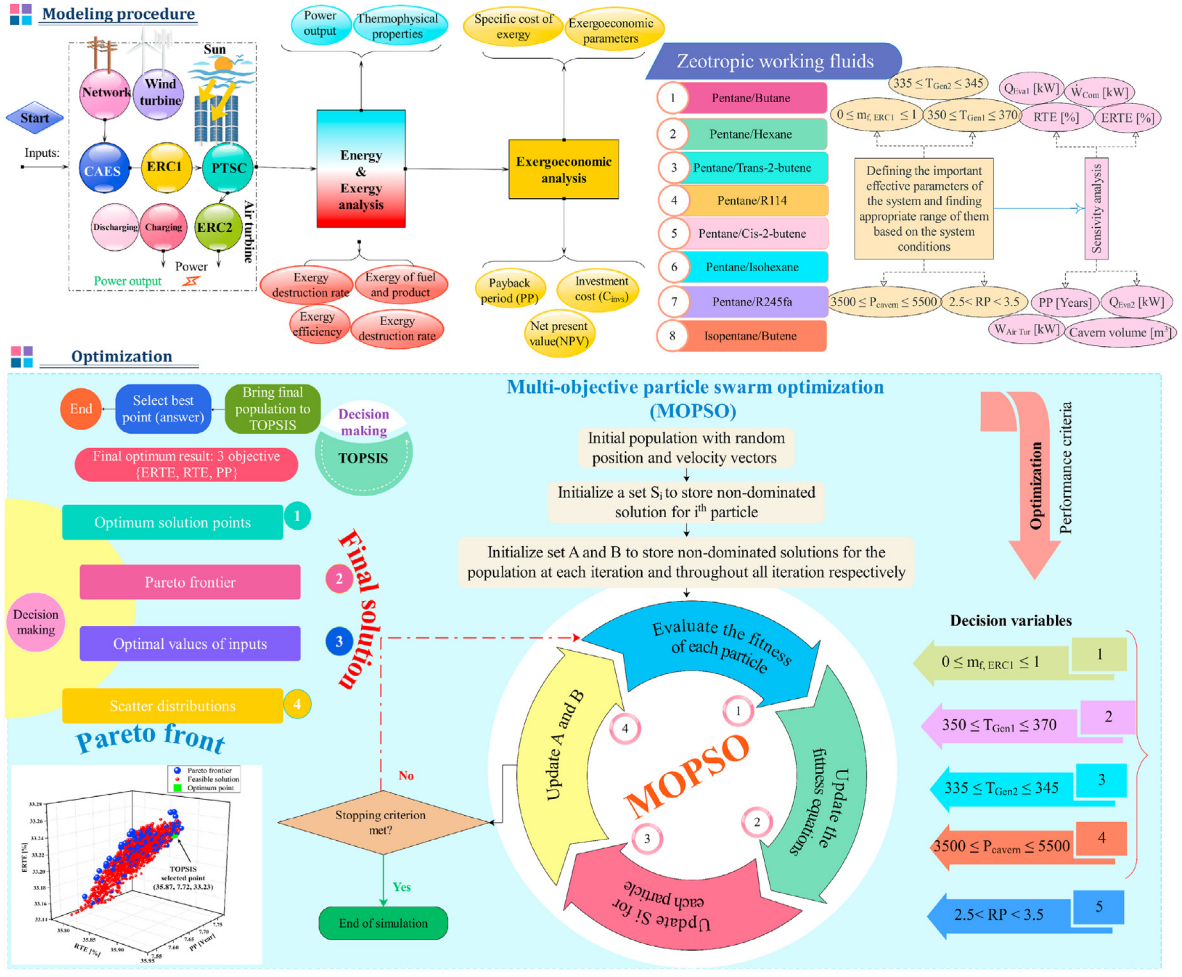


Fig. 1. The flowchart of the current study.

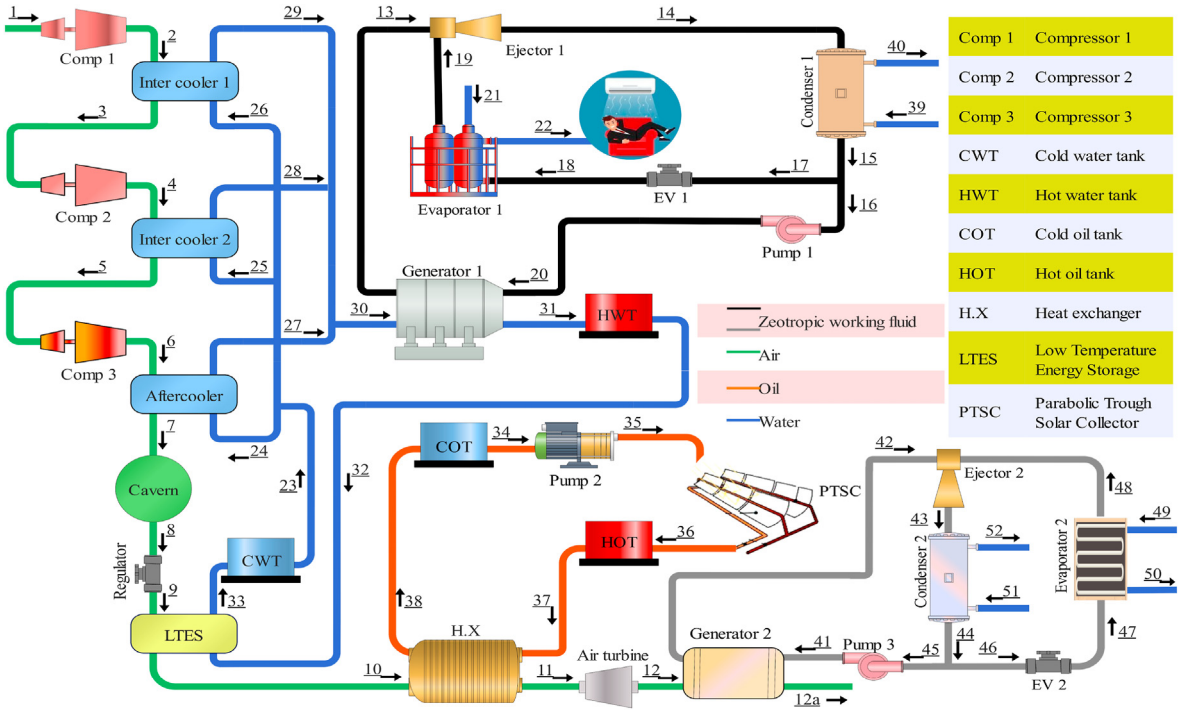
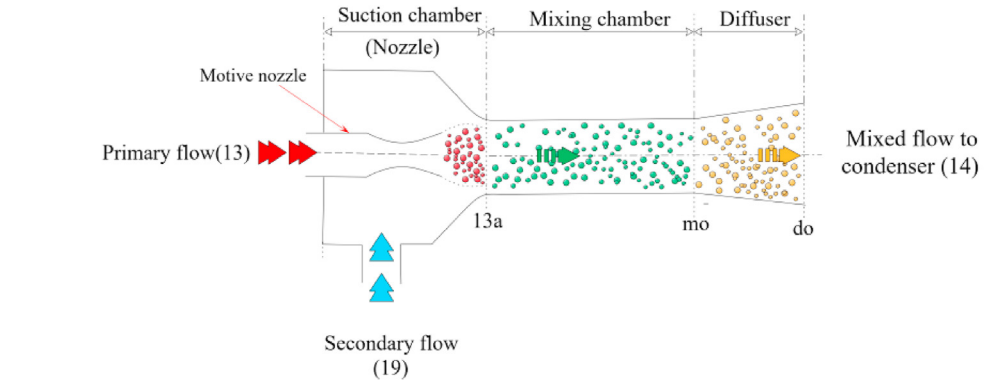


Fig. 2. The schematic diagram of the proposed cogeneration system.



(a)

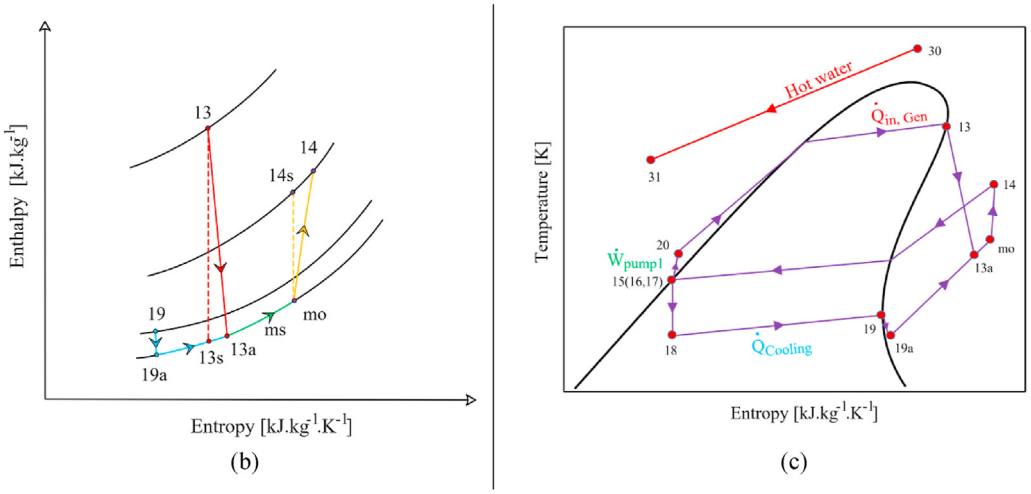


Fig. 3. The plot of the (a) ejector structure (b) h−s diagram of the ejector working processes (c) T-s diagram of the ERC cycle.

the solar system with Therminol working fluid collects the solar energy. Therminol is discharged from the cold tank and injected into the solar pump (state 34) to increase the pressure up to the solar collector's desired pressure. Then, the solar collector (state 35) is referred to absorb the solar energy. The hot Therminol is stored in the hot tank (state 36). At the grid peak time, the stored compressed air leaves the cavern (state 8) and, after passing through the pressure regulator valve, enters the low-temperature energy storage (LTES) unit (state 9). The hot tank's stored cooling water is sent to the LTES unit (state 32). The cooling water transfers heat to the compressed air in this unit and goes into the cold tank (state 33). The compressed air leaves the LTES and gets warmer; it is then delivered to the heat exchanger. It reaches the air turbine's desired inlet temperature by receiving energy from the stored hot Therminol discharged from the hot tank (state 37) at the grid peak time. After transferring the heat, Therminol is stored in the solar system cold tank (state 38), and the compressed air is injected into the air turbine (state 11). The hot compressed air is expanded inside the air turbine, and the extra power is produced to supply the grid's peak power demand. The air turbine exit flow is hot, so another ERC subsystem with the zeotropic mixture is added to the proposed system. The turbine exit flow before bringing to ambient supplies the second ERC system's inlet energy (state 12). In the second ERC subsystem, the condenser 2 outlet flow is divided into two streams (states 45 and 46). Stream 46, after crossing EV 2, enters evaporator 2 at the two-phase state (state 47). The zeotropic mixture produces a cooling load in evaporator 2 and enters ejector 2 as the secondary flow (state 48). Besides, stream 45 is pressurized in pump 3 and enters generator 2 to receive the air turbine's exit flow heat (state 42). At the saturated vapor state, the zeotropic mixture enters the ejector as the primary flow (state 42) and mixes with state 48. Then, the ejector outlet flow enters condenser 2 to cool down and complete the cycle.

In this work, various organic fluids are selected to form zeotropic mixtures in the proposed ERC performance. The thermodynamic properties of the organic fluids are presented in Table 1.

3. Mathematical modeling

The proposed system is analyzed by considering the following assumptions:

1. System components operate in steady-state conditions [40].
2. The energy and exergy rates of the kinetic and potential terms are insignificant [41].
3. The piping network and heat transfer-based components work with negligible pressure drop [42].
4. The air properties are calculated regarding ideal gas properties [43].
5. All components are adiabatically designed [44].
6. All turbines and pumps work at certain isentropic efficiencies [45].
7. The pressure regulator and throttling valve operate at an isenthalpic or constant enthalpy process [46].

Table 1
The physical properties of employed organic working fluids in the proposed system [39].

Fluid	T_{cr}[K]	P_{cr}[kPa]	MM [g.mol^{-1}]
Pentane	469.7	3370	72.15
Butane	425.13	3796	58.12
Trans-2-butene	428.61	4027	56.11
Cis-2-butene	435.75	4226	56.11
Hexane	507.82	3034	86.18
Isohexane	497.7	3040	86.18
R245fa	427.16	3651	134.05
Isopentane	460.35	3378	72.15
Butene	419.29	4005.1	56.11
R114	418.83	3257	170.92
Propane	369.89	4251.2	44.10
R115	353.1	3129	154.47

Accordingly, the input data required for modeling are listed in Table 2.

The modeling section deals with introducing the governed equations of the thermodynamic and exergoeconomic analyses and also simulating the ejector device, PTSCs, and CAES unit.

3.1. First law of thermodynamics

Concerning the first law of thermodynamics, Eqs. (1) and (2) individually illustrate the mass rate balance and energy rate balance equations for a control volume generally [56].

$$\sum \dot{m}_{in} = \sum \dot{m}_{out} \tag{1}$$

$$\dot{Q} - \dot{W} = \sum \dot{m}_{out} h_{out} - \sum \dot{m}_{in} h_{in} \tag{2}$$

In these equations, $\dot{m}$ denotes the mass flow rate of the employed working fluid, $\dot{Q}$ and $\dot{W}$ respectively are the heat transfer rate and work, and h stands for the specific enthalpy. Likewise, subscripts *in* and *out* refer to inlet and outlet.

3.2. Simulation of ejector

Since the proposed system contains an ejector, the ejector modeling is comprehensively discussed. Huang et al. [57] applied the 1-D constant-pressure mixing method to simulate an ejector. The primary flow enters the nozzle section of the ejector, and its pressure decreases with an increase in its velocity. Thereby, the energy conservation relation for the nozzle section can be written as [57]:

$$h_{pf,n1} + \frac{u_{pf,n_1}^2}{2} = h_{pf,n2} + \frac{u_{pf,n_2}^2}{2} \tag{3}$$

Here, the specific enthalpy of the primary flow at the nozzle's inlet and outlet is represented by h_{pf,n_1} and h_{pf,n_2}, correspondingly. In these conditions, the velocity of the primary flow at the inlet and outlet of the nozzle is shown by u_{pf,n_1} and u_{pf,n_2}, separately.

Since the primary flow's velocity at the inlet section of the nozzle is very low; hence, it can be ignored. Accordingly, the outlet velocity of the nozzle section is as follows [57]:

$$u_{pf,n_2} = \sqrt{2\left(h_{pf,n_1} - h_{pf,n_2}\right)} = \sqrt{2\eta_n\left(h_{pf,n_1} - h_{pf,is,n_2}\right)} \tag{4}$$

where η_n is a symbol to specify the nozzle's efficiency. This parameter has the following relation [57]:

$$\eta_n = \frac{h_{pf,n_1} - h_{pf,n_2}}{h_{pf,n_1} - h_{pf,is,n_2}} \tag{5}$$

Also, the energy balance and momentum balance equations in the mixing part of the ejector device are as follows [57]:

Table 2
Input data and initial assumptions to model the proposed system in this study.

Parameter	symbol	Value	unit	Ref.
Reference temperature	T_0	293.15	K	[47]
Reference pressure	P_0	101	kPa	[48]
Charging period	$t_{Charging}$	4	hr	[49]
Charging mass flow rate	$\dot{m}_{Charging}$	5	$kg.s^{-1}$	[50]
Charging pressure	P_{Cavern}	4500	kPa	[50]
Discharging period	$t_{Discharge}$	6	hr	[49]
Charging period to discharging period time ratio	TR1	0.67	–	[49]
Collector charging period to discharging period time ratio	TR2	1.67	–	–
Discharging mass flow rate of CAES	$\dot{m}_{Discharge}$	3.33	$kg.s^{-1}$	[49]
Discharging pressure	P_9	1500	kPa	[50]
LTES effectiveness	ε_{LTES}	70	%	[50]
Intercooler effectiveness	$\varepsilon_{Intercooler}$	80	%	[50]
Aftercooler effectiveness	$\varepsilon_{Aftercooler}$	80	%	[50]
Terminal temperature difference of heat exchanger	$\Delta T_{TT,\ H.X}$	10	K	[51]
Compressors isentropic efficiency	$\eta_{is,\ Comp}$	85.	%	[52]
Air turbine's isentropic efficiency	$\eta_{is,\ Air\ Tur}$	85.	%	[50]
Pumps isentropic efficiency	$\eta_{is,\ Pump}$	85.	%	[53]
The maximum to the minimum pressure ratio	PR	3	–	[50]
Specific heat of air at constant pressure	$C_{p,\ Air}$	1.01	$kJ.kg^{-1}.K^{-1}$	[50]
Specific heat of air at constant volume	$C_{v,\ Air}$	0.72	$kJ.kg^{-1}.K^{-1}$	[50]
Cooling water temperature	$T_{23}, T_{24}, T_{25}, T_{26}$	293.15.	K	[53]
Cooling water pressure	$P_{23}, P_{24}, P_{25}, P_{26}$	600.	kPa	–
Pinch point temperature difference of the condenser	$\Delta T_{PP,\ Cond}$	10	K	[54]
Pinch point temperature difference of the generator	$\Delta T_{PP,\ Gen}$	10.	K	[54]
Nozzle efficiency	η_n	90	%	[55]
Diffuser efficiency	η_d	90	%	[55]
Mixing chamber efficiency	η_m	90	%	[55]
Evaporator outlet temperature	T_{19}	280	K	[55]
Terminal temperature difference of the evaporator	$\Delta T_{TT,\ Eva}$	5	K	[55]
Single collector width	w	5.76	m	[54]
Single collector length	L	12.27	m	[54]
Receiver inner diameter	$D_{r,\ i}$	0.066	m	[54]
Receiver outer diameter	$D_{r,o}$	0.07	m	[54]
Cover inner diameter	$D_{c,\ i}$	0.115	m	[54]
Cover outer diameter	$D_{c,o}$	0.121	m	[54]
Emittance of the cover	ε_c	0.86	–	[54]
Emittance of the receiver	ε_r	0.15	–	[54]
The reflectance of the mirror	ρ_c	0.94	–	[54]
Intercept factor	γ	0.93	–	[54]
The transmittance of the glass cover	τ	0.96	–	[54]
The absorbance of the receiver	α	0.96	–	[54]
Incidence angle modifier	K_γ	1	–	[54]
Direct radiation heat	DNI	1000	$W.m^{-2}$	[54]

$$\dot{m}_{pf}\left(h_{pf,n1}+\frac{u_{pf,n_1}^2}{2}\right)+\dot{m}_{sf}\left(h_{sf,n2}+\frac{u_{sf,n_2}^2}{2}\right)=\left(\dot{m}_{pf}+\dot{m}_{sf}\right)\times\left(h_{mix}+\frac{u_{mix}^2}{2}\right) \tag{6}$$

$$\dot{m}_{pf}u_{pf,n1}+\dot{m}_{sf}u_{sf,n2}=\left(\dot{m}_{pf}+\dot{m}_{sf}\right)u_{mix,id} \tag{7}$$

where $u_{mix,id}$ denotes the mixed flow's velocity. It is worth mentioning that this velocity is computed without considering the energy dissipations in the mixing section.

On the other hand, as the secondary flow's velocity of the outlet of the nozzle section ($u_{sf,n2}$) can be negligible compared to the primary flow's velocity of the outlet of the nozzle section ($u_{pf,n2}$); thus, the entrainment ratio (μ) for the ejector is defined as [57]:

$$\mu=\dot{m}_{sf}\Big/m_{pf} \tag{8}$$

Subsequently, the momentum equation in the mixing section has the following relation [57]:

$$\mu=\frac{u_{pf,n_2}}{1+\mu} \tag{9}$$

Regarding the friction loss in the mixing section, the mixing efficiency is considered to assess the mixed flow's velocity as [57]:

$$\eta_m=\left(\frac{u_{mix}}{u_{mix,id}}\right)^2 \tag{10}$$

$$u_{mix}=\frac{u_{pf,n_2}}{1+\mu}\sqrt{\eta_m} \tag{11}$$

where u_{mix} is the mixed flow velocity with energy dissipation consideration. Consequently, the energy conservation equation, assuming the pressure drop neglected for the secondary flow, can be rewritten as follows [58]:

$$h_{mix} = \frac{h_{pf,n1} + \mu h_{sf,n_2}}{1+\mu} - \frac{u_{mix}^2}{2} \quad (12)$$

where $h_{sf,n1}$ signifies the specific enthalpy of the secondary flow at the inlet point.

The kinetic energy of the mixed flow is converted, and its pressure is augmented through the diffuser. With neglecting the diffuser's outlet velocity, the specific enthalpy of the outlet flow is presented as [57]:

$$h_{d2} = \frac{h_{pf,n1} + \mu h_{sf,n_2}}{1+\mu} \quad (13)$$

Here, the influence of the diffuser is considered via the diffuser efficiency (η_d) with the following equation:

$$\eta_d = \frac{h_{is,d_2} - h_{mix}}{h_{d_2} - h_{mix}} \quad (14)$$

On the other hand, the outlet specific enthalpy of the diffuser section under the ideal condition can be found through [57]:

$$h_{is,d_2} = h_{mix} + \eta_d (h_{d_2} - h_{mix}) \quad (15)$$

Finally, the entrainment ratio of the ejector device is formulated as [57]:

$$\mu = \sqrt{\eta_n \eta_m \eta_d \left(h_{pf,n_1} - h_{pf,is,n_2}\right) / \left(h_{is,d_2} - h_{mix}\right)} - 1 \quad (16)$$

3.3. Simulation of solar collectors

In this subsection, the modeling procedure and governed relations of the PTSC are presented. The below equation computes the absorbed solar irradiation by a PTSC unit [31,32]:

$$S = DNI.\eta_r \quad (17)$$

Herein, DNI denotes the direct normal irradiation and η_r indicates the efficiency of the PTSC receiver, which can be calculated by Refs. [59,60]:

$$\eta_r = \rho_c.\gamma.\tau.\alpha_r.k_\theta \quad (18)$$

In this equation, ρ_c is the mirror's reflectance factor; γ denotes the intercept coefficient; τ signifies the glass cover's transmittance; α_r is the receiver's absorbance, and k_θ indicates the incidence angle modifier with the following equation [59,60]:

$$k_\theta = 1 - 2.2307E-4\theta - 1.1E-4\theta^2 + 3.1860E-6\theta^3 - 4.8551E-8\theta^4 \quad (19)$$

Here, E depicts the time function, and θ denotes the incident angle and is calculated by Refs. [59,60]:

$$\cos(\theta) = \sqrt{\cos^2(\theta_z) + \cos^2(\delta).\sin^2(\omega)} \quad (20)$$

Herein, θ_z, δ, and ω are the zenith, declination, and hour angles, correspondingly.

The PTSC's useful power can be computed as follows [59,60]:

$$\dot{Q}_u = A_{ap}.F_R.\left(S - \left[\frac{A_r}{A_{ap}}\right].U_L.\left[T_{r,i} - T_0\right]\right) \quad (21)$$

where A and F_R are correspondingly the area and the heat removal factor. Also, U_L denotes the total heat loss coefficient of the PTSC unit. Subscripts of ap r i, and 0 stand for the aperture area, receiver, inlet, and ambient, respectively. The aperture and receiver areas are computed as [59,60]:

$$A_{ap} = L.(W - D_{c,o}) \quad (22)$$

$$A_r = \pi.D_{r,o}.L \quad (23)$$

In these relations, W is the width of the PTSC unit, L is its length, and D is its diameter. Moreover, subscripts c and o denote the cover and outlet, respectively.

The heat removal factor can be computed by Refs. [59,60]:

$$F_R = \left(\frac{\dot{m}_r.C_{pr}}{A_r.U_L}\right)\left(1 - e^{-\frac{A_r.U_L.F_1}{\dot{m}_r.C_{pr}}}\right) \quad (24)$$

where $\dot{m}_r$ denotes the receiver's oil mass flow rate. Also, F_1 indicates the performance factor of the PTSC unit that is calculated through [59,60]:

$$F_1 = \frac{U_O}{U_L} \quad (25)$$

in which, U_O denotes the total heat transfer coefficient of the PTSC unit. Also, U_L is the heat loss coefficient between the ambient and receiver and can be presented as [59,60]:

$$U_L = \frac{1}{\frac{A_r}{[h_{c,ca}+h_{r,ca}].A_c} + \frac{1}{h_{r,cr}}} \quad (26)$$

In this relation, $h_{c,ca}$ and $h_{r,ca}$ denote the convection and irradiation heat transfer coefficients between the cover and ambient, respectively. As well, $h_{r,cr}$ is the irradiation heat transfer coefficient between the cover and receiver. Furthermore, $h_{c,ca}$ has the following relation [59,60]:

$$h_{c,ca} = \frac{Nu.k_{air}}{D_{c,o}} \quad (27)$$

Here, Nu shows the Nusselt number and k_{air} denotes the thermal conductivity coefficient of the air.

Accordingly, $h_{r,ca}$ is presented as [59,60]:

$$h_{r,ca} = \varepsilon_{cv}.\sigma.(T_C + T_0).\left(T_C^2 + T_0^2\right) \quad (28)$$

where ε_{cv} and σ are the emittance factor of the cover and Stefan Boltzmann constant.

Lastly, $h_{r,cr}$ can be computed by Refs. [59,60]:

$$h_{r,cr} = \frac{\sigma.(T_C + T_{r,av}).\left(T_C^2 + T_{r,av}^2\right)}{\frac{1}{\varepsilon_r} + \left(\frac{A_r}{A_c}\right)\left(\frac{1}{\varepsilon_{cv}} - 1\right)} \quad (29)$$

In which, $T_{r,av}$ and ε_r are the mean temperature and emittance of the receiver.

The cover area of the PTSC unit is obtained by Refs. [59,60]:

$$A_C = \pi.D_{c,o}.L \quad (30)$$

Also, U_O can be computed as follows [59,60]:

$$U_O = \frac{1}{\left(\left[\frac{1}{U_L}\right] + \left[\frac{D_{r,o}}{h_{c,r,in}.D_{r,i}}\right] + \left[\frac{D_{r,o}}{2K_r}.\ln\left[\frac{D_{r,o}}{D_{r,i}}\right]\right]\right)} \quad (31)$$

In which, K_r denotes the receiver's thermal conductivity and $h_{c,r,in}$ indicates the convection heat transfer coefficient within the receiver with the following relation [59,60]:

$$h_{c,r,in} = \frac{Nu_r.k_r}{D_{r,i}} \quad (32)$$

The mean temperature of the cover can be calculated by Refs. [59,60]:

$$T_C = \frac{h_{r,cr}.T_{r,av} + \left(\frac{A_c}{A_r}\right)(h_{c,ca} + h_{r,ca}).T_0}{h_{r,cr} + \left(\frac{A_c}{A_r}\right)(h_{c,ca} + h_{r,ca})} \quad (33)$$

The number of the PTSC units for meeting the demanded energy is given as follows [61]:

$$N_{PTSC} = \frac{(\dot{m}_{31}.h_{31} - \dot{m}_{37}.h_{37})}{\dot{Q}_u} \quad (34)$$

3.4. Simulation of compressed air energy storage

Generally, the input power for an air compressor can be obtained via the following relation [56]:

$$\dot{W}_{AC} = \dot{m}(h_{out} - h_{in}) \quad (35)$$

Also, the isentropic efficiency for a compressor is defined as follows [56]:

$$\eta_{AC} = \frac{h_{out,s} - h_{in}}{h_{out} - h_{in}} \quad (36)$$

Subsequently, as the current study uses three air compressors, the overall power consumed by these devices has the following equation:

$$\dot{W}_{AC} = \dot{W}_{AC_1} + \dot{W}_{AC_2} + \dot{W}_{AC_3} \quad (37)$$

The volume of the air storage vessel depends on the mass flow rate of the charging and discharging periods and air density, which is defined as [30]:

$$V_{cavern} = \frac{3600\dot{m}_7 t_{ch}}{(\rho_{7-}\rho_9)} = \frac{3600\dot{m}_8 t_{dch}}{(\rho_{7-}\rho_9)} \quad (38)$$

Here, t_{ch} shows the charging period and t_{dch} indicates the discharging period.

The air turbine of the CAES unit generates power taking into account the following relation [56]:

$$\dot{W}_{AT} = \dot{m}(h_{in} - h_{out}) \quad (39)$$

Finally, the isentropic efficiency of the mentioned air turbine is [56]:

$$\eta_{AT} = \frac{(h_{in} - h_{out})}{(h_{in} - h_{out,s})} \quad (40)$$

3.5. Second law of thermodynamics

Exergy as an influential tool divided from the second law of thermodynamics is recognized as an appropriate approach for analyzing the quality of the processes. Hence, the exergy rate balance at the steady-state conditions for a control volume is given by Ref. [56]:

$$\dot{E}_Q - \dot{W} = \sum \dot{m}_{out} e_{out} - \sum \dot{m}_{in} e_{in} + \dot{E}_D \quad (41)$$

where $\dot{E}_D$ determines the exergy destruction rate. Also, $\dot{E}_Q$ is a symbol to show the exergy rate of a heat transfer process to the ambient based on the following relation [62]:

$$\dot{E}_Q = \sum \dot{Q}\left(1 - \frac{T_0}{T}\right) \quad (42)$$

Here, T denotes the temperature of the boundary where heat transfer has occurred. As well, T_0 refers to the dead state or reference state temperature. The specific physical exergy flow for a stream can be presented as follows [62]:

$$e = h - h_0 - T_0(s - s_0) \quad (43)$$

In this relation, s denotes the specific entropy.

Furthermore, the mass rate, energy rate, and also exergy rate balances for each component of the whole integrated system are listed in Tables 3 and 4.

3.6. Exergoeconomic analysis

Exergy and energy analyses represent the thermodynamically statement (first and second laws of thermodynamics) for the performance of energy conversion-based systems. However, the energy and exergy approaches do not illustrate a perfect definition for the mentioned systems; therefore, an economic analysis is required to complete the performance analysis of such systems. In this regard, the exergoeconomic analysis as a useful method to evaluate the financial aspect of a system based on its exergy is utilized in the current research. Accordingly, the cost rate balance equation for a control volume can be expressed as [63]:

$$\dot{C}_P = \dot{C}_F + \dot{C}^{CL} + \dot{C}^{OM} \quad (44)$$

where;

$$\dot{C} = c\dot{E} \quad (45)$$

$\dot{C}_P$ is the total cost rate of products, $\dot{C}_F$ is the total cost rate of the fuels, $\dot{C}^{CL}$ is the fixed cost associated with the fixed investment cost rate, and $\dot{C}^{OM}$ is the operation and maintenance cost rate. In order to evaluate the unit exergy cost of the input and output flows ($\dot{E}_{in}$ and $\dot{E}_{out}$), work or power ($\dot{E}_w$), and heat transfer ($\dot{E}_q$) terms, the following equations are utilized [63]:

$$\dot{C}_{in} = c_{in}\dot{E}_{in} = c_{in}(\dot{m}_{in}e_{in}) \quad (46)$$

$$\dot{C}_{out} = c_{out}\dot{E}_{out} = c_{out}(\dot{m}_{out}e_{out}) \quad (47)$$

$$\dot{C}_w = c_w\dot{E}_w = c_w(\dot{m}_w e_w) \quad (48)$$

$$\dot{C}_q = c_q\dot{E}_q = c_q(\dot{m}_q e_q) \quad (49)$$

Here, c shows the unit exergy cost.

Table 3
Mass and energy rate balance for the components of the proposed system in this study.

Component	Mass balance	Energy balance
Compressor 1	$\dot{m}_1 = \dot{m}_2$	$\dot{W}_{Comp1} = \dot{m}_2 h_2 - \dot{m}_1 h_1 = (\dot{m}_2 h_{2s} - \dot{m}_1 h_1)/\eta_{is,Comp}$
Compressor 2	$\dot{m}_3 = \dot{m}_4$	$\dot{W}_{Comp2} = \dot{m}_4 h_4 - \dot{m}_3 h_3 = (\dot{m}_4 h_{4s} - \dot{m}_3 h_3)/\eta_{is,Comp}$
Compressor 3	$\dot{m}_5 = \dot{m}_6$	$\dot{W}_{Comp3} = \dot{m}_6 h_6 - \dot{m}_5 h_5 = (\dot{m}_6 h_{6s} - \dot{m}_5 h_5)/\eta_{is,Comp}$
Intercooler 1	$\dot{m}_2 = \dot{m}_3, \dot{m}_{26} = \dot{m}_{29}$	$\dot{Q}_{Cooler1} = \dot{m}_2 h_2 - \dot{m}_3 h_3 = \dot{m}_{29} h_{29} - \dot{m}_{26} h_{26}$
Intercooler 2	$\dot{m}_4 = \dot{m}_5, \dot{m}_{25} = \dot{m}_{28}$	$\dot{Q}_{Cooler2} = \dot{m}_4 h_4 - \dot{m}_5 h_5 = \dot{m}_{28} h_{28} - \dot{m}_{25} h_{25}$
Aftercooler	$\dot{m}_6 = \dot{m}_7, \dot{m}_{24} = \dot{m}_{27}$	$\dot{Q}_{Cooler3} = \dot{m}_6 h_6 - \dot{m}_7 h_7 = \dot{m}_{27} h_{27} - \dot{m}_{24} h_{24}$
Hot water tank	$\dot{m}_{32} = TR1 \times \dot{m}_{31}$	$\dot{Q}_{HWT}^{Loss} = \dot{m}_{32}(h_{31} - h_{32})$
Cold water tank	$TR1 \times \dot{m}_{23} = \dot{m}_{33}$	$\dot{Q}_{CWT}^{Loss} = \dot{m}_{23}(h_{33} - h_{23})$
Cavern	$\dot{m}_8 = TR1 \times \dot{m}_7$	$\dot{Q}_{Cavern}^{Loss} = \dot{m}_8(h_7 - h_8)$
Regulator	$\dot{m}_8 = \dot{m}_9$	$\dot{m}_8 h_8 = \dot{m}_9 h_9$
LTES	$\dot{m}_9 = \dot{m}_{10}, \dot{m}_{32} = \dot{m}_{33}$	$\dot{Q}_{LTES} = \dot{m}_{10} h_{10} - \dot{m}_9 h_9 = \dot{m}_{32} h_{32} - \dot{m}_{33} h_{33}$
Heat exchanger	$\dot{m}_{10} = \dot{m}_{11}, \dot{m}_{37} = \dot{m}_{38}$	$\dot{Q}_{H.X} = \dot{m}_{11} h_{11} - \dot{m}_{10} h_{10} = \dot{m}_{37} h_{37} - \dot{m}_{38} h_{38}$
Air turbine	$\dot{m}_{11} = \dot{m}_{12}$	$\dot{W}_{Air\ Tur} = \dot{m}_{11} h_{11} - \dot{m}_{12} h_{12} = (\dot{m}_{11} h_{11} - \dot{m}_{12} h_{12s})\eta_{is,Air\ Tur}$
Collector	$\dot{m}_{35} = \dot{m}_{36}$	$\dot{Q}_u = N_{Coll}.A_{ap}.F_R.\left(S - \left[\frac{A_r}{A_{ap}}\right].U_L.[T_{r,i} - T_0]\right)$
Pump 2	$\dot{m}_{34} = \dot{m}_{35}$	$\dot{W}_{Pump2} = \dot{m}_{35} h_{35} - \dot{m}_{34} h_{34} = (\dot{m}_{35} h_{35s} - \dot{m}_{34} h_{34})/\eta_{is,Pump}$
Hot oil tank	$\dot{m}_{37} = TR2 \times \dot{m}_{36}$	$\dot{Q}_{HOT}^{Loss} = \dot{m}_{37}(h_{36} - h_{37})$
Cold oil tank	$TR2 \times \dot{m}_{34} = \dot{m}_{38}$	$\dot{Q}_{COT}^{Loss} = \dot{m}_{34}(h_{38} - h_{34})$
Generator 1	$\dot{m}_{30} = \dot{m}_{31}, \dot{m}_{13} = \dot{m}_{20}$	$\dot{Q}_{Gen1} = \dot{m}_{30} h_{30} - \dot{m}_{31} h_{31} = \dot{m}_{13} h_{13} - \dot{m}_{20} h_{20}$
Ejector 1	$\dot{m}_{14} = \dot{m}_{13} + \dot{m}_{19}$	$\dot{m}_{14} h_{14} = \dot{m}_{13} h_{13} + \dot{m}_{19} h_{19}$
Condenser 1	$\dot{m}_{14} = \dot{m}_{15}, \dot{m}_{39} = \dot{m}_{40}$	$\dot{Q}_{Cond1} = \dot{m}_{14} h_{14} - \dot{m}_{15} h_{15} = \dot{m}_{40} h_{40} - \dot{m}_{39} h_{39}$
Expansion valve 1	$\dot{m}_{17} = \dot{m}_{18}$	$\dot{m}_{17} h_{17} = \dot{m}_{18} h_{18}$
Evaporator 1	$\dot{m}_{18} = \dot{m}_{19}, \dot{m}_{21} = \dot{m}_{22}$	$\dot{Q}_{Eva1} = \dot{m}_{19} h_{19} - \dot{m}_{18} h_{18} = \dot{m}_{21} h_{21} - \dot{m}_{22} h_{22}$
Pump 1	$\dot{m}_{16} = \dot{m}_{20}$	$\dot{W}_{Pump1} = \dot{m}_{20} h_{20} - \dot{m}_{16} h_{16} = (\dot{m}_{20} h_{20s} - \dot{m}_{16} h_{16})/\eta_{is,\ Pump}$
Generator 2	$\dot{m}_{12} = \dot{m}_{12a}, \dot{m}_{41} = \dot{m}_{42}$	$\dot{Q}_{Gen2} = \dot{m}_{12} h_{12} - \dot{m}_{12a} h_{12a} = \dot{m}_{42} h_{42} - \dot{m}_{41} h_{41}$
Ejector 2	$\dot{m}_{43} = \dot{m}_{42} + \dot{m}_{48}$	$\dot{m}_{43} h_{43} = \dot{m}_{42} h_{42} + \dot{m}_{48} h_{48}$
Condenser 2	$\dot{m}_{43} = \dot{m}_{44}, \dot{m}_{51} = \dot{m}_{52}$	$\dot{Q}_{Cond2} = \dot{m}_{43} h_{43} - \dot{m}_{44} h_{44} = \dot{m}_{52} h_{52} - \dot{m}_{51} h_{51}$
Expansion valve 2	$\dot{m}_{46} = \dot{m}_{47}$	$\dot{m}_{46} h_{46} = \dot{m}_{47} h_{47}$
Evaporator 2	$\dot{m}_{47} = \dot{m}_{48}, \dot{m}_{49} = \dot{m}_{50}$	$\dot{Q}_{Eva2} = \dot{m}_{48} h_{48} - \dot{m}_{47} h_{47} = \dot{m}_{49} h_{49} - \dot{m}_{50} h_{50}$
Pump 3	$\dot{m}_{41} = \dot{m}_{45}$	$\dot{W}_{Pump3} = \dot{m}_{41} h_{41} - \dot{m}_{45} h_{45} = (\dot{m}_{41} h_{41s} - \dot{m}_{45} h_{45})/\eta_{is,\ Pump}$

Table 4
The exergy balance of the employed components.

Component	Fuel Exergy	product Exergy	Exergy destruction
Compressor 1	$\dot{E}_F^{Comp1} = \dot{W}_{Comp1}$	$\dot{E}_P^{Comp1} = (\dot{E}_2 - \dot{E}_1)$	$\dot{E}_D^{Comp1} = \dot{E}_F^{Comp1} - \dot{E}_P^{Comp1}$
Compressor 2	$\dot{E}_F^{Comp2} = \dot{W}_{Comp2}$	$\dot{E}_P^{Comp2} = (\dot{E}_4 - \dot{E}_3)$	$\dot{E}_D^{Comp2} = \dot{E}_F^{Comp2} - \dot{E}_P^{Comp2}$
Compressor 3	$\dot{E}_F^{Comp3} = \dot{W}_{Comp3}$	$\dot{E}_P^{Comp3} = (\dot{E}_6 - \dot{E}_5)$	$\dot{E}_D^{Comp3} = \dot{E}_F^{Comp3} - \dot{E}_P^{Comp3}$
Intercooler 1	$\dot{E}_F^{Cooler1} = (\dot{E}_2 - \dot{E}_3)$	$\dot{E}_P^{Cooler1} = (\dot{E}_{29} - \dot{E}_{26})$	$\dot{E}_D^{Cooler1} = \dot{E}_F^{Cooler1} - \dot{E}_P^{Cooler1}$
Intercooler 2	$\dot{E}_F^{Cooler2} = (\dot{E}_4 - \dot{E}_5)$	$\dot{E}_P^{Cooler2} = (\dot{E}_{28} - \dot{E}_{25})$	$\dot{E}_D^{Cooler2} = \dot{E}_F^{Cooler2} - \dot{E}_P^{Cooler2}$
Aftercooler	$\dot{E}_F^{Cooler3} = (\dot{E}_6 - \dot{E}_7)$	$\dot{E}_P^{Cooler3} = (\dot{E}_{27} - \dot{E}_{24})$	$\dot{E}_D^{Cooler3} = \dot{E}_F^{Cooler3} - \dot{E}_P^{Cooler3}$
Hot water tank	$\dot{E}_F^{HWT} = \dot{E}_{31}$	$\dot{E}_P^{HWT} = \dot{E}_{32}$	$\dot{E}_D^{HWT} = \dot{E}_F^{HWT} - \dot{E}_P^{HWT}$
Cold water tank	$\dot{E}_F^{CWT} = \dot{E}_{33}$	$\dot{E}_P^{CWT} = \dot{E}_{23}$	$\dot{E}_D^{CWT} = \dot{E}_F^{CWT} - \dot{E}_P^{CWT}$
Cavern	$\dot{E}_F^{Cavern} = \dot{E}_7$	$\dot{E}_P^{Cavern} = \dot{E}_8$	$\dot{E}_D^{Cavern} = \dot{E}_F^{Cavern} - \dot{E}_P^{Cavern}$
Regulator	$\dot{E}_F^{Reg} = \dot{E}_8$	$\dot{E}_P^{Reg} = \dot{E}_9$	$\dot{E}_D^{Reg} = \dot{E}_F^{Reg} - \dot{E}_P^{Reg}$
LTES	$\dot{E}_F^{LTES} = (\dot{E}_{32} - \dot{E}_{33})$	$\dot{E}_P^{LTES} = (\dot{E}_{10} - \dot{E}_9)$	$\dot{E}_D^{LTES} = \dot{E}_F^{LTES} - \dot{E}_P^{LTES}$
Heat exchanger	$\dot{E}_F^{H.X} = (\dot{E}_{37} - \dot{E}_{38})$	$\dot{E}_P^{H.X} = (\dot{E}_{11} - \dot{E}_{10})$	$\dot{E}_D^{H.X} = \dot{E}_F^{H.X} - \dot{E}_P^{H.X}$
Air turbine	$\dot{E}_F^{Air\ Tur} = (\dot{E}_{11} - \dot{E}_{12})$	$\dot{E}_P^{Air\ Tur} = \dot{W}_{Air\ Tur}$	$\dot{E}_D^{Air\ Tur} = \dot{E}_F^{Air\ Tur} - \dot{E}_P^{Air\ Tur}$
Collector	$\dot{E}_F^{Coll} = N_{Coll}.A_{ap}.DNI\left(1 + \frac{1}{3}\left(\frac{T_0}{T_{sun}}\right)^4 - \frac{4}{3}\left(\frac{T_0}{T_s}\right)\right)$	$\dot{E}_P^{Coll} = (\dot{E}_{36} - \dot{E}_{35})$	$\dot{E}_D^{Coll} = \dot{E}_F^{Coll} - \dot{E}_P^{Coll}$
Pump 2	$\dot{E}_F^{Pump2} = \dot{W}_{Pump2}$	$\dot{E}_P^{Pump2} = (\dot{E}_{35} - \dot{E}_{34})$	$\dot{E}_D^{Pump2} = \dot{E}_F^{Pump2} - \dot{E}_P^{Pump2}$
Hot oil tank	$\dot{E}_F^{HOT} = \dot{E}_{36}$	$\dot{E}_P^{HOT} = \dot{E}_{37}$	$\dot{E}_D^{HOT} = \dot{E}_F^{HOT} - \dot{E}_P^{HOT}$
Cold oil tank	$\dot{E}_F^{COT} = \dot{E}_{38}$	$\dot{E}_P^{COT} = \dot{E}_{34}$	$\dot{E}_D^{COT} = \dot{E}_F^{COT} - \dot{E}_P^{COT}$
Generator 1	$\dot{E}_F^{Gen1} = (\dot{E}_{30} - \dot{E}_{31})$	$\dot{E}_P^{Gen1} = (\dot{E}_{13} - \dot{E}_{20})$	$\dot{E}_D^{Gen1} = \dot{E}_F^{Gen1} - \dot{E}_P^{Gen1}$
Ejector 1	$\dot{E}_F^{Eje1} = (\dot{E}_{13} + \dot{E}_{19})$	$\dot{E}_P^{Eje1} = \dot{E}_{14}$	$\dot{E}_D^{Eje1} = \dot{E}_F^{Eje1} - \dot{E}_P^{Eje1}$
Condenser 1	$\dot{E}_F^{Cond1} = (\dot{E}_{14} - \dot{E}_{15})$	$\dot{E}_P^{Cond1} = (\dot{E}_{40} - \dot{E}_{39})$	$\dot{E}_D^{Cond1} = \dot{E}_F^{Cond1} - \dot{E}_P^{Cond1}$
Expansion valve 1	$\dot{E}_F^{EV1} = \dot{E}_{17}$	$\dot{E}_P^{EV1} = \dot{E}_{18}$	$\dot{E}_D^{EV1} = \dot{E}_F^{EV1} - \dot{E}_P^{EV1}$

Table 4 (*continued*)

Component	Fuel Exergy	product Exergy	Exergy destruction
Evaporator 1	$\dot{E}_F^{Eva1} = (\dot{E}_{18} - \dot{E}_{19})$	$\dot{E}_P^{Eva1} = (\dot{E}_{22} - \dot{E}_{21})$	$\dot{E}_D^{Eva1} = \dot{E}_F^{Eva1} - \dot{E}_P^{Eva1}$
Pump 1	$\dot{E}_F^{Pump1} = \dot{W}_{Pump1}$	$\dot{E}_P^{Pump1} = (\dot{E}_{20} - \dot{E}_{16})$	$\dot{E}_D^{Pump1} = \dot{E}_F^{Pump1} - \dot{E}_P^{Pump1}$
Generator 2	$\dot{E}_F^{Gen2} = (\dot{E}_{12} - \dot{E}_{12a})$	$\dot{E}_P^{Gen2} = (\dot{E}_{42} - \dot{E}_{41})$	$\dot{E}_D^{Gen2} = \dot{E}_F^{Gen2} - \dot{E}_P^{Gen2}$
Ejector 2	$\dot{E}_F^{Eje2} = (\dot{E}_{42} + \dot{E}_{48})$	$\dot{E}_P^{Eje2} = \dot{E}_{43}$	$\dot{E}_D^{Eje2} = \dot{E}_F^{Eje2} - \dot{E}_P^{Eje2}$
Condenser 2	$\dot{E}_F^{Cond2} = (\dot{E}_{43} - \dot{E}_{44})$	$\dot{E}_P^{Cond2} = (\dot{E}_{52} - \dot{E}_{51})$	$\dot{E}_D^{Cond2} = \dot{E}_F^{Cond2} - \dot{E}_P^{Cond2}$
Expansion valve 2	$\dot{E}_F^{EV2} = \dot{E}_{46}$	$\dot{E}_P^{EV2} = \dot{E}_{47}$	$\dot{E}_D^{EV2} = \dot{E}_F^{EV2} - \dot{E}_P^{EV2}$
Evaporator 2	$\dot{E}_F^{Eva2} = (\dot{E}_{47} - \dot{E}_{48})$	$\dot{E}_P^{Eva2} = (\dot{E}_{50} - \dot{E}_{49})$	$\dot{E}_D^{Eva2} = \dot{E}_F^{Eva2} - \dot{E}_P^{Eva2}$
Pump 3	$\dot{E}_F^{Pump3} = \dot{W}_{Pump3}$	$\dot{E}_P^{Pump3} = (\dot{E}_{41} - \dot{E}_{45})$	$\dot{E}_D^{Pump3} = \dot{E}_F^{Pump3} - \dot{E}_P^{Pump3}$

Table 5 presents the cost balance equations and relative auxiliary equations for the components of the suggested system in this work.

Also, the exergy destruction cost rate of component i can be expressed as [63]:

$$\dot{C}_{D,i} = c_{P,i}\dot{E}_{D,i} (\text{if } \dot{E}_{F,i} = \text{constant value}) \tag{50}$$

On the other hand, the total investment cost rate of component i is calculated as the sum of the capital cost rate ($\dot{Z}^{CL}$) and the operation and maintenance cost rate ($\dot{Z}^{OM}$) [63]:

$$\dot{Z}_i = \dot{Z}^{CL} + \dot{Z}^{OM} \tag{51}$$

Accordingly, the total investment cost rate of component i can be evaluated as [63]:

$$\dot{Z}_i = \frac{CRF \times \varphi}{N \times 3600} \times Z_i \tag{52}$$

where Z_i is the purchase cost of component i, N is the annual runtime of the system, φ is the maintenance factor and is equal to 1.06, and CRF is the capital recovery factor, which can be calculated from the following equation [63]:

$$CRF = \frac{K(1+K)^n}{(1+K)^n - 1} \tag{53}$$

where K is the interest rate and assumed 0.15 and n is the expected lifetime for the component of the proposed system and assumed 20 years.

All equations related to the investment cost functions of the components employed in the system shown in Fig. 2 are listed in Table 6. Concerning the following equation, all investment cost functions should be updated to the target year [63].

$$\text{Cost of target year} = \text{cost of reference yare} \times \frac{\text{cost index of target year}}{\text{cost index of reference year}} \tag{54}$$

Subsequently, the fixed cost (FC) of the whole system is calculated [64]:

$$FC = C_{invs} + C_{inst} \tag{55}$$

where C_{invs} is the investment cost and C_{inst} is the installation and piping costs of the system. Consequently, these costs are equal to Ref. [64]:

$$C_{invs} = \sum_i Z_i \tag{56}$$

$$C_{inst} = 0.1 \times C_{invs} \tag{57}$$

Besides, the operating cost (OC) of the whole system can be calculated from Ref. [64]:

$$OC = 0.06 \times FC \tag{58}$$

In this regard, the net present value (NPV) and payback period (PP) investigations of the whole system are as below [64]:

$$PP = \frac{FC}{AS} \tag{59}$$

Table 5
Cost rate balance and auxiliary equations required for each component of the proposed system in this study.

Component	Cost balance	Auxiliary equations
Compressor 1	$\dot{C}_1 + \dot{Z}_{Comp1} + \dot{C}_{W,Comp1} = \dot{C}_2$	$c_1 = 0$
Compressor 2	$\dot{C}_3 + \dot{Z}_{Comp2} + \dot{C}_{W,Comp2} = \dot{C}_4$	–
Compressor 3	$\dot{C}_5 + \dot{Z}_{Comp3} + \dot{C}_{W,Comp3} = \dot{C}_6$	–
Intercooler 1	$\dot{C}_2 + \dot{C}_{26} + \dot{Z}_{Cooler1} = \dot{C}_3 + \dot{C}_{29}$	$c_2 = c_3, c_{26} = c_{23}$
Intercooler 2	$\dot{C}_4 + \dot{C}_{25} + \dot{Z}_{Cooler2} = \dot{C}_5 + \dot{C}_{28}$	$c_4 = c_5, c_{25} = c_{23}$
Aftercooler	$\dot{C}_6 + \dot{C}_{24} + \dot{Z}_{Cooler3} = \dot{C}_7 + \dot{C}_{27}$	$c_6 = c_7, c_{24} = c_{23}$
Hot water tank	$\dot{C}_{31} + \dot{Z}_{HWT} = \dot{C}_{32}$	–
Cold water tank	$\dot{C}_{33} + \dot{Z}_{CWT} = \dot{C}_{23}$	–
Cavern	$\dot{C}_7 + \dot{Z}_{Cavern} = \dot{C}_8$	–
Regulator	$\dot{C}_8 + \dot{Z}_{Reg} = \dot{C}_9$	–
LTES	$\dot{C}_9 + \dot{C}_{32} + \dot{Z}_{LTES} = \dot{C}_{10} + \dot{C}_{33}$	$c_9 = c_{10}$
Heat exchanger	$\dot{C}_{10} + \dot{C}_{37} + \dot{Z}_{H.X} = \dot{C}_{11} + \dot{C}_{38}$	$c_{37} = c_{38}$
Air turbine	$\dot{C}_{11} + \dot{Z}_{Air\ Tur} = \dot{C}_{12} + \dot{C}_{W,Air\ Tur}$	$c_{11} = c_{12}$
Collector	$\dot{C}_{35} + \dot{Z}_{Coll} + \dot{C}_{Sun} = \dot{C}_{36}$	$c_{Sun} = 0$
Pump 2	$\dot{C}_{34} + \dot{Z}_{Pump2} + \dot{C}_{W,Pump2} = \dot{C}_{35}$	–
Hot oil tank	$\dot{C}_{36} + \dot{Z}_{HOT} = \dot{C}_{37}$	–
Cold oil tank	$\dot{C}_{38} + \dot{Z}_{COT} = \dot{C}_{34}$	–
Generator 1	$\dot{C}_{30} + \dot{C}_{20} + \dot{Z}_{Gen1} = \dot{C}_{13} + \dot{C}_{31}$	$c_{30} = c_{31}$
Ejector 1	$\dot{C}_{13} + \dot{C}_{19} + \dot{Z}_{Eje1} = \dot{C}_{14}$	–
Condenser 1	$\dot{C}_{14} + \dot{C}_{39} + \dot{Z}_{Cond1} = \dot{C}_{15} + \dot{C}_{40}$	$c_{39} = 0, c_{14} = c_{15}$
Expansion valve 1	$\dot{C}_{17} + \dot{Z}_{EV1} = \dot{C}_{18}$	–
Evaporator 1	$\dot{C}_{18} + \dot{C}_{21} + \dot{Z}_{Eva1} = \dot{C}_{19} + \dot{C}_{22}$	$c_{21} = 0, c_{19} = c_{18}$
Pump 1	$\dot{C}_{16} + \dot{Z}_{Pump1} + \dot{C}_{W,Pump1} = \dot{C}_{20}$	–
Generator 2	$\dot{C}_{12} + \dot{C}_{41} + \dot{Z}_{Gen2} = \dot{C}_{12a} + \dot{C}_{42}$	$c_{12} = c_{12a}$
Ejector 2	$\dot{C}_{42} + \dot{C}_{48} + \dot{Z}_{Eje2} = \dot{C}_{43}$	–
Condenser 2	$\dot{C}_{43} + \dot{C}_{51} + \dot{Z}_{Cond2} = \dot{C}_{44} + \dot{C}_{52}$	$c_{51} = 0, c_{43} = c_{44}$
Expansion valve 2	$\dot{C}_{46} + \dot{Z}_{EV2} = \dot{C}_{47}$	–
Evaporator 2	$\dot{C}_{47} + \dot{C}_{49} + \dot{Z}_{Eva2} = \dot{C}_{48} + \dot{C}_{50}$	$c_{49} = 0, c_{47} = c_{48}$
Pump 3	$\dot{C}_{45} + \dot{Z}_{Pump3} + \dot{C}_{W,Pump3} = \dot{C}_{41}$	–

Table 6
The cost function for each component of the proposed system in this study.

Components	Cost functions	Ref. year	Cost index	Reference
Compressor 1	$Z_{Comp1} = 7900(\dot{W}_{Comp1})^{0.62}$	2009	521.9	[50]
Compressor 2	$Z_{Comp2} = 7900(\dot{W}_{Comp2})^{0.62}$	2009	521.9	[50]
Compressor 3	$Z_{Comp3} = 7900(\dot{W}_{Comp3})^{0.62}$	2009	521.9	[50]
Intercooler 1	$Z_{Cooler1} = 309.14(A_{Cooler1})^{0.85}$	2001	394.3	[65]
Intercooler 2	$Z_{Cooler2} = 309.14(A_{Cooler2})^{0.85}$	2001	394.3	[65]
Aftercooler	$Z_{Cooler3} = 309.14(A_{Cooler3})^{0.85}$	2001	394.3	[65]
Hot water tank	$Z_{HWT} = 5941.7(V_{HWT})^{-0.272}$	2000	394.1	[66]
Cold water tank	$Z_{CWT} = 5941.7(V_{HWT})^{-0.389}$	2000	394.1	[66]
Cavern	$Z_{Cavern} = 1.218 f_m.\exp(2.631 + 1.3673(\ln V_{Cavern}) - 0.06309(\ln V_{Cavern})^2)$, $f_m = 1$(Carbon steel)	2001	394.3	[27]
Regulator	$Z_{Reg} = 114.5 \times \dot{m}_8$	2000	394.1	[33]
LTES	$Z_{LTES} = 309.14(A_{LTES})^{0.85}$	2001	394.3	[65]
Heat exchanger	$Z_{H.X} = 309.14(A_{H.X})^{0.85}$	2001	394.3	[65]
Air turbine	$Z_{Air\ Tur} = 1100(\dot{W}_{Air\ Tur})^{0.81}$	2009	521.9	[50]
Collector	$Z_{coll} = 170 \times A_{ap}$	2018	638.1	[67]
Pump 2	$Z_{Pump2} = 705.48(\dot{W}_{Pump2})^{0.71}.\left[1 + \left(\frac{0.2}{1-\eta_{is,Pump}}\right)\right]$	2018	638.1	[67]
Hot oil tank	$Z_{HOT} = 423(V_{HOT})$	2018	638.1	[68]
Cold oil tank	$Z_{COT} = 423(V_{COT})$	2018	638.1	[68]
Generator 1	$Z_{Gen} = 309.14(A_{Gen})^{0.85}$	2001	394.3	[65]
Ejector 1	$Z_{Eje} = 1000 \times 16.14 \times 0.989 \times \dot{m}_{13}\left(\frac{T_{13}}{0.1P_{13}}\right)^{0.05} \times (0.1P_{14})^{-0.75}$	2001	394.3	[55]
Condenser 1	$Z_{Cond1} = 516.62(A_{Cond})^{0.6}$	2001	394.3	[65]
Expansion valve 1	$Z_{EV} = 114.5 \times \dot{m}_{17}$	2000	394.1	[66]
Evaporator 1	$Z_{Eva} = 309.14(A_{Eva})^{0.85}$	2001	394.3	[65]
Pump 2	$Z_{Pump1} = 200(\dot{W}_{Pump1})^{0.65}$	2001	394.3	[55]
Generator 2	$Z_{Gen2} = 309.14(A_{Gen2})^{0.85}$	2001	394.3	[65]
Ejector 2	$Z_{Eje2} = 1000 \times 16.14 \times 0.989 \times \dot{m}_{42}\left(\frac{T_{42}}{0.1P_{42}}\right)^{0.05} \times (0.1P_{43})^{-0.75}$	2001	394.3	[55]
Condenser 2	$Z_{Cond2} = 516.62(A_{Cond2})^{0.6}$	2001	394.3	[65]
Expansion valve 2	$Z_{EV2} = 114.5 \times \dot{m}_{46}$	2000	394.1	[66]
Evaporator 2	$Z_{Eva2} = 309.14(A_{Eva2})^{0.85}$	2001	394.3	[65]
Pump 3	$Z_{Pump3} = 200(\dot{W}_{Pump3})^{0.65}$	2001	394.3	[55]

$$NPV = -(FC \times IF_0 \times RDF_0) + \sum_{i=1}^{n}(AS \times IF_i \times RDF_i) \quad (60)$$

where AS is the annual net saving money, IF_i is the inflation factor, and RDF_i is the real discount factor. The mentioned factors have the following relations [64]:

$$AS = AI - OC \quad (61)$$

$$IF_i = \left(1 + \frac{R}{100}\right)^{-i} \quad (62)$$

$$RDF_i = \left(1 + \frac{RIR}{100}\right)^{-i} \quad (63)$$

Here, i shows the year number, AI depicts the total annual income, R signifies the inflation rate (5%), and RIR designates the real interest rate (3%).

3.7. Performance criteria

Since the proposed system operates in charging and discharging periods, the conventional energy and exergy efficiencies cannot be applied to evaluate the system's performance. In this regard, RTE and ERTE. The RTE is defined as the ratio of total output energy (electricity and cooling load) to total input energy (compressors and pumps input power + input solar energy). Thus:

$$RTE = \frac{(t_{ch} \times \dot{Q}_{Eva1}) + (t_{dch} \times (\dot{W}_{Air\,tur} + \dot{Q}_{Eva2})}{t_{ch} \times (\dot{W}_{AC} + \dot{W}_{Pump\,1}) + t_{dch} \times (\dot{Q}_{Coll} + \dot{W}_{Pump\,2} + \dot{W}_{Pump\,3})} \quad (64)$$

Also, the ERTE can be defined as the ratio of output exergy to input exergy, which can be determined as:

$$ERTE = \frac{\left(t_{ch} \times \dot{Q}_{Eva1}\left[1 - \frac{T_0}{T_{Eva1}}\right]\right) + \left(t_{dch} \times \left(\dot{W}_{Airtur} + \dot{Q}_{Eva2}\left[1 - \frac{T_0}{T_{Eva2}}\right]\right)\right.}{t_{ch} \times (\dot{W}_{AC} + \dot{W}_{Pump1}) + t_{dch} \times \left(\dot{E}_F^{Coll} + \dot{W}_{Pump2} + \dot{W}_{Pump3}\right)} \quad (65)$$

4. Result and discussion

This section presents the study results in four main parts, including working fluid selection for ERC 1 subsystems, main results, parametric study results, and net present value estimation.

4.1. Zeotropic working fluid selection for ERC 1 subsystem

In this subsection, the zeotropic mixture's mass fraction variation's effect on the ejector's entertainment ratio, cooling capacity, coefficient of performance, evaporator 1 and vapor generator 1's glide temperature, power consumption of the ERC 1's pump, round trip efficiency (RTE), exergetic round trip efficiency (ERTE), and the

payback period is studied. In this way, Pentane/Butane, Pentane/Hexane, Pentane/*Trans*-2-butene, Pentane/R114, Pentane/*Cis*-2-butene, Pentane/Isohexane, Pentane/R245fa, and Isopentane/Butene are select as the ERC 1 subsystem working fluid's candidates.

4.1.1. Ejector 1's entertainment ratio

The change in entrainment ratio of the ejector based on increasing the mass fraction of the zeotropic mixtures between 0 and 1 is plotted in Fig. 4 for all mentioned zeotropic working fluids. This plot reveals that there exists an optimal point (maximum value) for the entrainment ratio with increasing the mentioned mass fraction. Because in this situation, considering the thermo-physical properties of the mixtures, initially, the mass fraction of the secondary sucked into the ejector increases and overcame the primary flow's mass fraction, while in higher mass fractions, the mass flow rate of the secondary flow experiences a decline, decreasing the entrainment ratio. Nevertheless, the Pentane/Butane, Pentane/R245fa, and Pentane/*Trans*-2-butane zeotropic mixtures achieve the highest entertainment ratio of 0.222 at their optimal state among the others.

4.1.2. Cooling capacity

In Fig. 5, the effect of increasing the mass fraction of the evaluated zeotropic working fluids in the ERC subsystem on the generated cooling load is demonstrated. According to this figure, the cooling load ascends to the maximum point and then descends. As shown in Fig. 5, the Pentane/Butane, Isopentane/Butene, and Pentane/*Trans*-2-butane correspondingly represent the maximum cooing production capacity at 50-50, 0.63–0.37, and 60-40 mass fractions with a value of 163 kW, and the Pentane/Isohexane shows the minimum cooling load at a mass fraction of 30–70.

4.1.3. Coefficient of performance

Fig. 6 represents the variation in the COP of the ERC versus the change in mass fraction of the zeotropic working fluids. Referring to this plot, it is evident that the COP also has an increasing-decreasing behavior, which is completely affected by the behavior of the generated cooling load. Because the COP has a direct relation with the cooling production capacity of the refrigeration cycle; hence, the highest COP belongs to the Pentane/*Trans*-2-butane working fluid with a value of 0.163 and Pentane/Butane of about 0.16, while the Pentane/Isohexane has the lowest COP of 0.142 both in their optimal state.

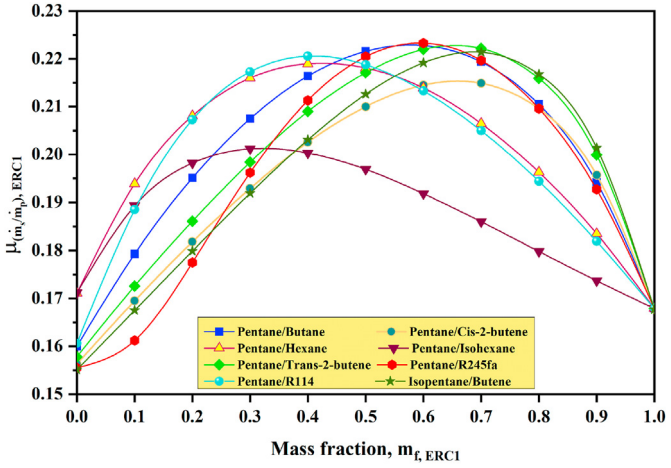


Fig. 4. The effect of zeotropic mixtures' mass fraction on the ejector 1's entertainment ratio.

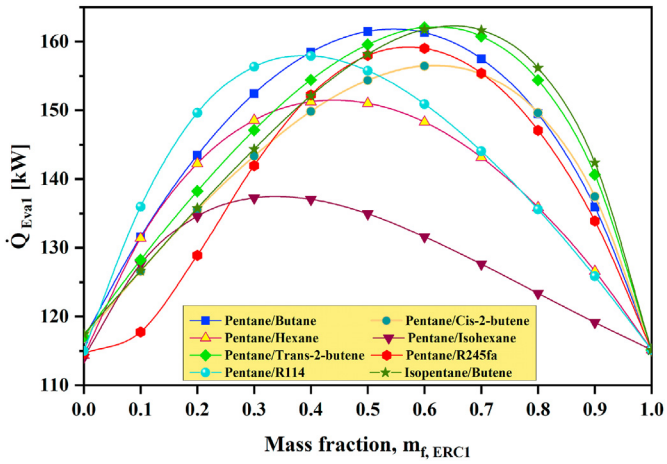


Fig. 5. The effect of zeotropic mixtures' mass fraction on the cooling capacity.

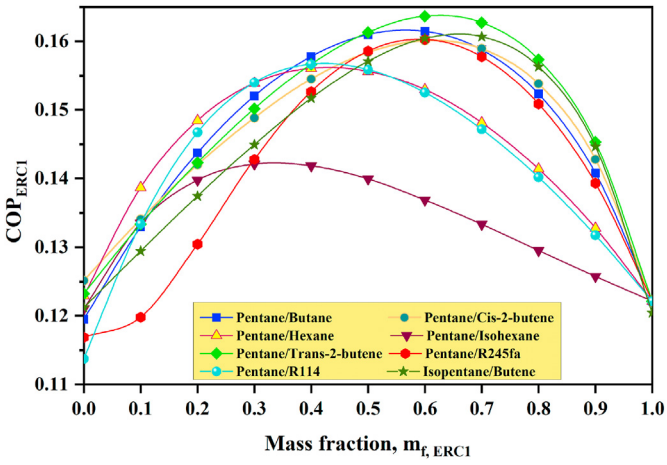


Fig. 6. The effect of zeotropic mixtures' mass fraction on the ERC subsystem COP.

4.1.4. Evaporator 1 and vapor generator 1's glide temperature

The ERC 1 subsystem uses a vapor generator and an evaporator as heat exchangers; therefore, the temperature glide of these components is recognized as another performance indicator of the ERC 1 unit, which has a strong impact on the mentioned components. In this way, Figs. 7 and 8 individually show the influence of the mass fraction of the zeotropic mixtures on the temperature glide of evaporator 1 and vapor generator 1. These figures reveal that the temperature glide initially augments and then declines. Also, the zeotropic mixtures with the highest COP reach the highest temperature glide of the mentioned components in their optimal mode. According to the results attained from these figures, the Pentane/Butane and Pentane/*Trans*-2-butane represent the highest temperature glide in both evaporator 1 and vapor generator 1. Nevertheless, the maximum temperature glide of the evaporator based on using Pentane/Butane and Pentane/*Trans*-2-butane is 9.2 K and 9.0 K, respectively (see Fig. 7). Likewise, for the steam generator, the maximum temperature glide of the mentioned working fluids is 8.8 K (see Fig. 8).

4.1.5. ERC 1's pump power consumption

With respect to Fig. 9, pump 1's power consumption capacity is illustrated by increasing the mass fraction of the selected zeotropic working fluids. There are different behaviors for the selected

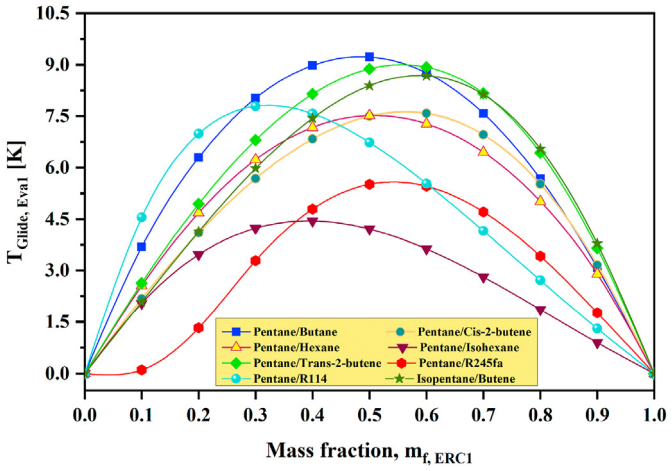


Fig. 7. The effect of zeotropic mixtures' mass fraction on the evaporator 1's glide temperature.

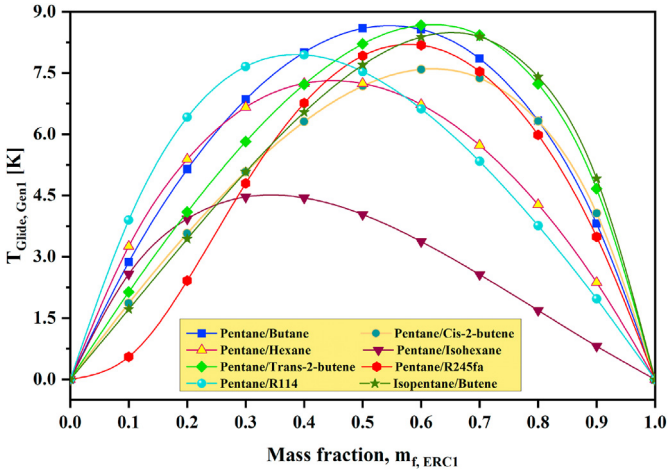


Fig. 8. The effect of zeotropic mixtures' mass fraction on the vapor generator 1's glide temperature.

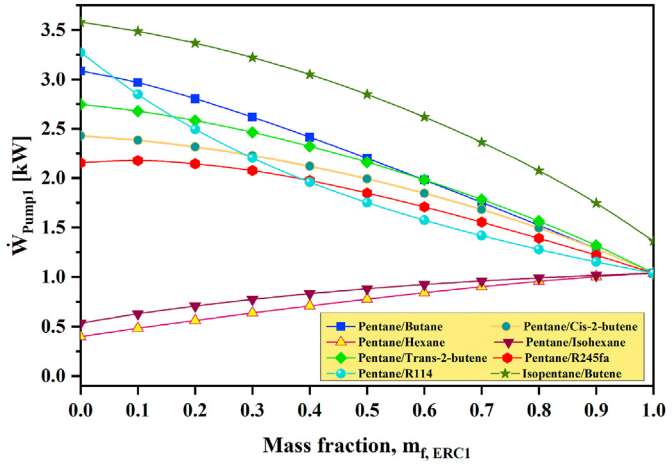


Fig. 9. The effect of zeotropic mixtures' mass fraction on the ERC pump's power consumption.

working fluids based on their thermo-physical features. However, the obtained results indicate that the Pentane/Hexane requires the minimum and Isopentane/Butene requires the maximum input power of pump 1.

4.1.6. Round trip and exergetic round trip efficiencies

As previously mentioned, the RTE and ERTE indicate the overall performance of the CAES system. Thereby, the effect of utilizing different zeotropic working fluids with regard to the variation in their mass fraction on the RTE and ERTE is studied and represented in Figs. 10 and 11, respectively. Considering the cooling generation behavior with increasing the mass fraction, the observed trends for the RTE and ERTE are justifiable. Fig. 10 specifies that the Pentane/Butane and Pentane-Trans-2-butane show the highest RTE at the mass fractions of 50-50 and 60–40 compared to the other selected groups of zeotropic working fluids, respectively. In this way, the maximum RTE obtained for the mentioned working fluids is around 36.2%. On the other hand, the obtained results for the ETRE demonstrate different results. The cooling capacity has a minor influence on the ETRE compared to net power production and exergy destruction. Since the net power production and system's inlet energy are assumed constant so, the exergy destruction has the main impact on the proposed system ERTE. In this way, the Pentane/Hexane and Pentane/Isohexane present the highest. But, the ETRE variation for all zeotropic mixtures is performed in the small divergence in which it varies between 33.15 and 33.23%, as shown in Fig. 11.

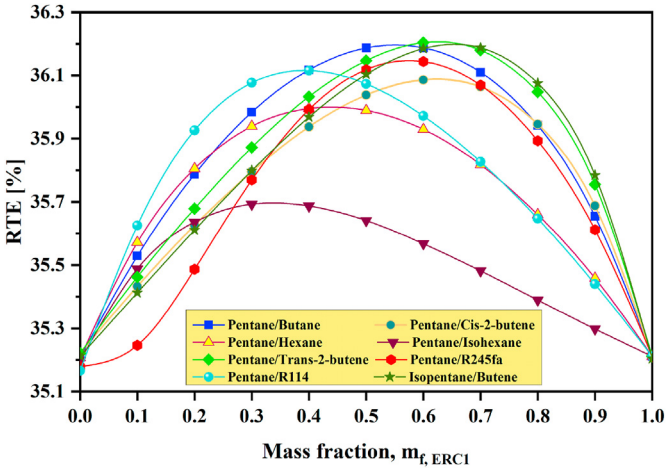


Fig. 10. The effect of zeotropic mixtures' mass fraction on the proposed system RTE.

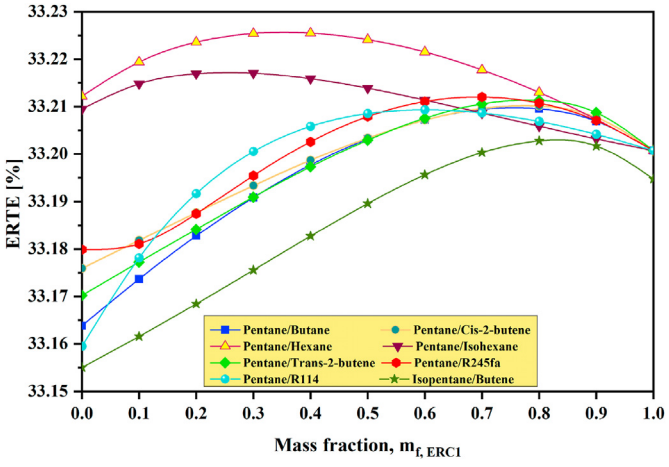


Fig. 11. The effect of zeotropic mixtures' mass fraction on the proposed system ERTE.

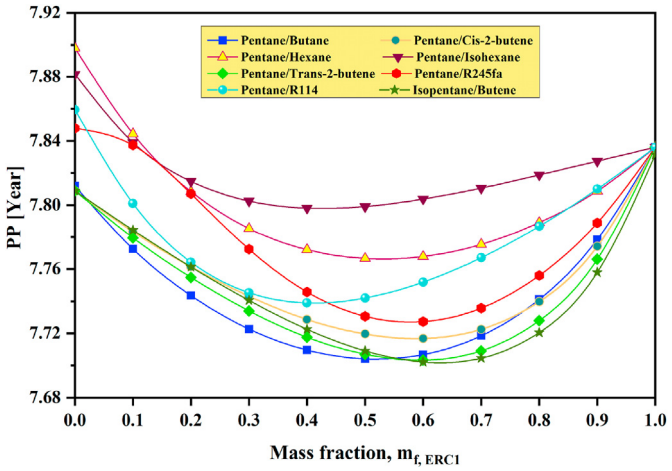


Fig. 12. The effect of zeotropic mixtures' mass fraction on the proposed system payback period.

4.1.7. Payback period

Fig. 12 displays the influence of the mass fraction of the selected working fluids on the system's payback period. Although this variation has a minor effect on the system's payback period due to the low cost of the refrigeration unit compared to the other subsystems; however, there is a downward-upward behavior; because the highest cooling load receives the highest income and reduces the payback period. As shown in Fig. 11, the Pentane/Butane and Isopentane/Butene represent the minimum payback period, and the Pentane/Isohexane represents the maximum payback period.

According to the comparative study to select the best working fluid, the effect of increasing the mass fraction of different zeotropic working fluids on various crucial variables of the system's performance is analyzed. Hence, the appropriate zeotropic working fluid with its optimum mass fraction is determined to employ in the ERC subsystem. In this way, the Pentane/Butane with a mass fraction of 50-50 is introduced as the best working fluid for the ERC unit, among the others. Also, the ERC 2 subsystem works based on a low-temperature heat source; hence, the Propane/R115 is selected as its working fluid. Propane/R115 has a low critical temperature that can be simply adjusted with low-temperature source. Also, Propane/R115 provides performance than the other zeotropic fluids.

4.2. Main results

After selecting a proper working fluid for ERC 1, the thermodynamic and thermo-economic analyses based on Tables 3–6 are applied to the proposed system. Table 7 contains the thermodynamic and exergoeconomic properties, including temperature, pressure, specific enthalpy, exergy rate, specific entropy, mass flow

Table 7
Thermodynamic and economic properties for each state of the proposed system at the base operational condition.

State	T [K]	P [kPa]	h [kJ.kg^{-1}]	s [kJ.kg^{-1}.K^{-1}]	ṁ[kg.s^{-1}]	c [$.GJ^{-1}]	$\dot{E}$ [kW]	$\dot{C}$ [$]
1	293.15	101	293.41	6.84	5	0	0	0
2	442.56	358.06	444.37	6.9	5	21.31	677848.2	363980
3	323.03	358.06	323.01	6.58	5	21.31	539443.7	289662
4	487.1	1269.35	489.54	6.63	5	20.87	1295006	681090
5	331.94	1269.35	330.39	6.24	5	20.87	1075650	565723
6	501.01	4500	502.53	6.29	5	20.64	1858884	966895
7	334.72	4500	327.84	5.87	5	20.64	1606690	835717
8	293.15	4500	283.32	5.72	3.33	21.1	1061533	564368
9	293.15	1500	290.12	6.06	3.33	29.65	755681.8	564631
10	332.25	1500	330.3	6.19	3.33	30.67	763883.7	590454
11	641.08	1500	650.92	6.87	3.33	22.24	1167231	654259
12	352.62	101	353.35	7.03	3.33	22.24	17892.71	10029
12a	336.52	101	337.1	6.98	3.33	22.24	9819.41	5504
13	360.18	682.86	576.62	1.98	2.25	72.66	172332.7	315555
14	339.71	196.26	554.58	2.06	2.75	65.55	82661.68	136539
15	303.94	196.26	131.4	0.7	2.75	65.55	16180.74	26727
16	303.94	196.26	131.4	0.7	2.25	65.55	13246.61	21880
17	303.94	196.26	131.4	0.7	0.5	65.55	2934.12	4847
18	270.77	57.19	131.4	0.72	0.5	715	271.17	4886
19	280	57.19	455.02	1.89	0.5	715	10095.6	181902
20	304.2	682.86	132.38	0.7	2.25	58.86	15130.72	22443
21	288	101	62.45	0.22	12.85	0	2462.09	0
22	285	101	49.88	0.18	12.85	1208	6212.29	189105
23	293.15	600	84.48	0.3	5.4	1618	2699.13	110024
24	293.15	600	84.48	0.3	1.8	1618	899.71	36675
25	293.15	600	84.48	0.3	1.8	1618	899.71	36675
26	293.15	600	84.48	0.3	1.8	1618	899.71	36675
27	408.57	600	569.73	1.69	1.8	48.73	138288.1	169824
28	398.44	600	526.56	1.58	1.8	52.2	117040	153948
29	373.64	600	421.59	1.31	1.8	62.54	71618.2	112875
30	393.59	600	505.96	1.53	5.4	53	326946.2	436647
31	349.61	600	320.61	1.03	5.4	53	109156.7	145782
32	349	600	318.05	1.03	3.6	54.5	71279.75	97900
33	340.12	600	280.85	0.92	3.6	54.5	53042.31	72851
34	344	13500	91.83	0.25	1.01	26.39	19257.05	12807
35	344.4	15000	93.56	0.25	1.01	26.95	20777.47	14109
36	651.08	15000	728.11	1.54	1.01	11.01	278540.4	77309
37	651.08	15000	728.11	1.54	1.68	4.919	464234	57548
38	344	15000	92.89	0.25	1.68	4.919	34462.41	4272

(*continued on next page*)

Table 7 (*continued*)

State	T [K]	P [kPa]	h [kJ.kg^{-1}]	s [kJ.kg^{-1}.K^{-1}]	ṁ[kg.s^{-1}]	c [$.GJ^{-1}]	$\dot{E}$ [kW]	$\dot{C}$ [$]
39	293.15	101	84.01	0.3	22.46	0	0	0
40	305.55	101	135.87	0.47	22.46	188.2	23958.53	113629
41	305.77	2473.3	268.72	1.28	0.26	73.04	19957.32	36734
42	336.13	2473.3	475.75	1.9	0.26	63.04	26158.03	41557
43	310.1	1249.5	470.82	1.95	0.31	73.71	25782.81	47894
44	304.5	1249.5	266.62	1.28	0.31	73.71	23316.63	43313
45	304.5	1249.5	266.62	1.28	0.26	73.71	19487.36	36200
46	304.5	1249.5	266.62	1.28	0.05	73.71	3829.27	7113
47	279.41	645.16	266.62	1.29	0.05	76.86	3674.79	7117
48	280	645.16	445.69	1.93	0.05	76.86	3232.04	6260
49	288	101	62.45	0.22	0.73	0	140.32	0
50	285	101	49.88	0.18	0.73	130.2	354.05	1161
51	293.15	101	84.01	0.3	2.45	0	0	0
52	299.38	101	110.08	0.38	2.45	328.5	669.89	5545

rate, unit exergy cost, and cost rate for each state of the whole application illustrated in Fig. 2.

On the other hand, Table 8 presents the results of crucial variables attained from the overall analysis. According to this table, at the baseline design conditions, the air turbine and ERCs unit produce 991.89 kW power and 170.7 kW cooling capacity. Here, the RTE and ERTE are found to be 35.82% and 33.19%. Lastly, the payback period of the system is calculated as 7.75 years.

Fig. 13 shows the exergy analysis results for the components utilized in the whole setup in both charging and discharging periods. At charging time, the solar collectors with an exergy destruction rate of 556.8 kW have a 39.79% share in the total exergy destruction rate, which is introduced as the main irreversibility source. Also, the ERC subsystem, with a value of 15.45% share in the total exergy destruction rate, is recognized as the second-worst irreversibility source. On the other hand, during the discharging period, with 60.19% and 30.98% values, the regulator valve and air turbine have the highest exergy destruction ratio amen the other components.

4.3. Parametric study

In this section, the effect of cavern inlet pressure, direct normal irradiation, and charging to discharging pressure ratio's variation on the system's performance criteria is investigated. In this way, the cooling capacity in the ERC 1 and 2, air turbine power generation, cavern volume, compressors' power consumption, RTE, ERTE, and payback period are considered as the performance criteria.

Table 8
Main performance metrics of the proposed system in this study.

Parameter	Present work
$\dot{W}_{Comp1}$ [kW]	754.79
$\dot{W}_{Comp2}$ [kW]	832.66
$\dot{W}_{Comp3}$ [kW]	860.69
$\dot{W}_{Air\ Tur}$ [kW]	991.89
$\dot{Q}_{Eva1}$ [kW]	161.5
COP_{ERC1} [−]	0.16
$\dot{Q}_{Eva2}$ [kW]	9.2
COP_{ERC2} [−]	0.11
$\dot{Q}_{Cooler1}$ [kW]	606.81
$\dot{Q}_{Cooler2}$ [kW]	795.75
$\dot{Q}_{Cooler3}$ [kW]	873.45
$\dot{Q}_{LTES}$ [kW]	133.92
Cavern volume [m^3]	2495.5
$\dot{E}_D^{Charging,\ Total}$ [kW]	1290.47
$\dot{E}_D^{Discharging.Total}$ [kW]	517.99
PP [Years]	7.75
RTE [%]	35.82
ERTE [%]	33.19
$\eta_{Energy,\ Coll}$ [%]	73.31
$\eta_{Exergy,\ Coll}$ [%]	31.64

4.3.1. The effect of cavern inlet pressure

The effect of cavern inlet pressure (charging pressure) increasing between 3500 and 5500 kPa on the system's performance criteria is shown in Fig. 14. This augment increases the stored air density, so the cavern volume reduces the constant inlet air mass flow rate. Also, the compressors' power consumption increases from 2240 kW to 2610 kW. Since the ERC 1 receives its inlet energy from rejected heat during the compressing process, subsequently, the ERC 1's cooling capacity raises, as shown in Fig. 14a. Besides, the air turbine's inlet temperature and pressure ratio are assumed constant, and these operating constraints influence the ERC 2 inlet energy, leading to a decline in its production cooling capacity. It reaches zero at 5500 kPa. The air turbine's power production enhances by increasing the cavern inlet from 926 kW to 1035 kW. Since the system's production rate is enhanced, the proposed system's RTE and ERTE improve from 35.46% to 36% and 32.6%–33.5%, respectively. The positive impact of the cavern inlet pressure on the system's products refers to a declining the payback period from 7.86 years to 7.66 years (see Fig. 14b).

4.3.2. The effect of direct normal irradiation

Fig. 15 presents the effect of DNI's variation on the proposed system's performance criteria. The DNI factor is ranged between 500 and 1200 for this parametric analysis since the system is available within this range of DNI. As shown in Fig. 15a, increasing the DNI does not influence the charging time operated components; hence, the compressors' power consumption, cavern volume, and ERC 1's cooling capacity remain constant. But, increasing the DNI improves the air turbine inlet flow's received energy and enhances the ERC 2's cooling capacity and air turbine's power production from 6.9 kW to 983 kW–9.5 kW and 994 kW, respectively. Therefore, the system's RTE and ERTE increased from 35.15% to 36% and 32.6%–33.3%. The products' generation enhancement declines the system's payback period. In this way, the proposed system's payback period reduces from 8.6 years to 7.6 years, which is shown in Fig. 15b.

4.3.3. The effect of charging to discharging pressure ratio (PR)

Increasing the PR is the consequence of decreasing the discharging pressure, and the charging pressure remains constant. Regarding the mentioned assumption, the compressors' power consumption and ERC 1's generated cooling load remain constant. The discharging pressure decline influences the cavern volume and

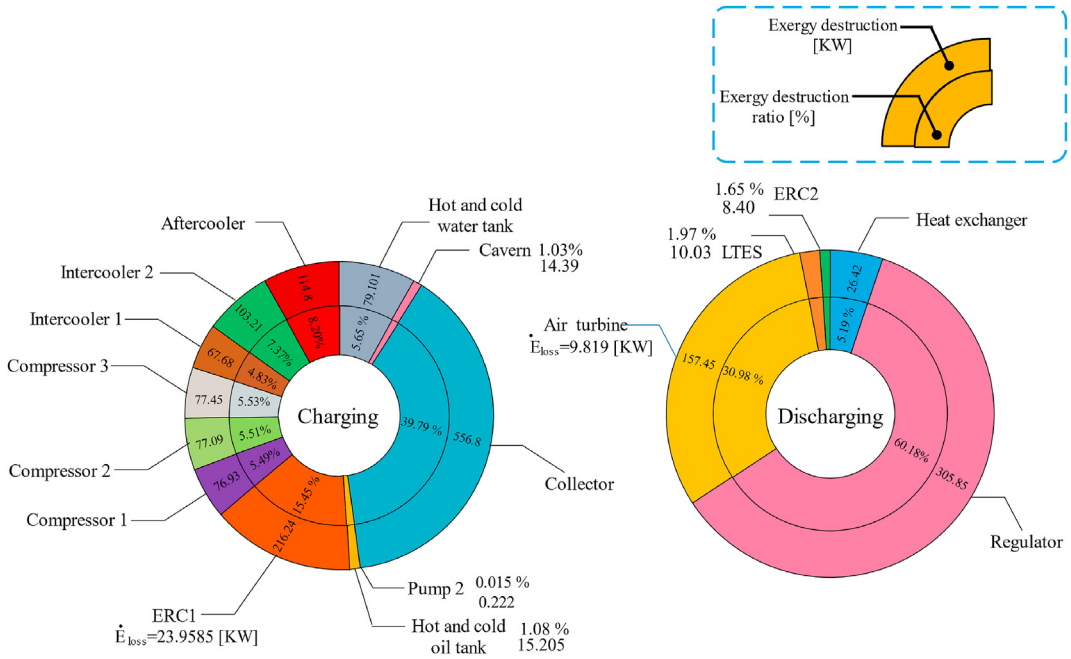


Fig. 13. Contribution of each component of the whole system to the exergy destruction rate and exergy destruction ratio.

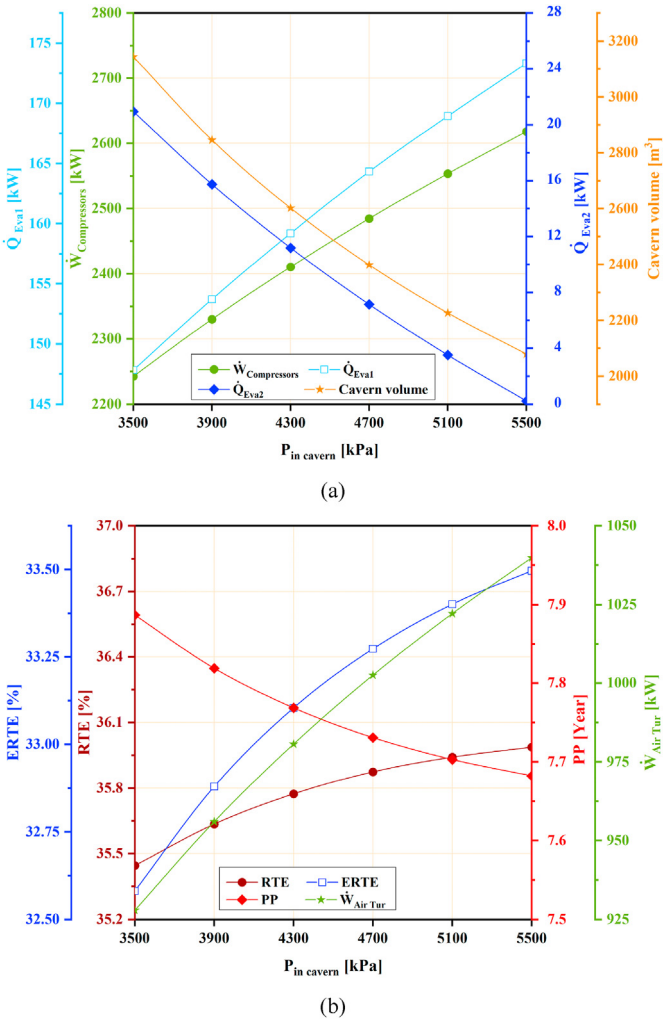


Fig. 14. The effect of cavern inlet pressure variation on the system's performance.

reduces it from 2850 m^3 to 2300 m^3 (see Fig. 16a). Also, this reduction directly affects the air turbine's operating condition and reduces its power production from 1040 kW to 952 kW. But the ERC 2's performance enhances, and its cooling capacity from 1 kW to 16 kW. Reducing the air turbine's power generation influences the system's RTE and ERTE higher than ERC 2's cooling capacity augment. So, the proposed system's RTE and ERTE witness descendant trends, and their values decrease from 37.5% to 34.7% and 34.7%–32%, respectively. Furthermore, the system's payback is affected by product reduction and increases from 7.3 years to 8.15 years, shown in Fig. 16b.

4.4. Net present value estimation

The net present value estimation provides useful data about the proposed system's economic performance during its designed lifetime. NPV is a strong criterion to determine if a project is profitable or not. A positive NPV implies that the financial position of the investor will be improved by undertaking the project. A negative NPV would indicate a financial loss. Zero or null NPV would mean that the present value of all benefits over the useful lifetime is equal to the present value of all the costs. In this way, this index helps the investors to plan an economic perspective of their investment. In this section, the proposed system's NPV is estimated for three different purchases, and product sale prices are plotted in Fig. 17. First scenario proposes 0.08 $\$.kWh^{-1}$ for purchase cost of input power [49], 0.318 $\$.kWh^{-1}$ for electricity [49] and 0.08 $\$.kWh^{-1}$ for cooling load sale prices [69]. In this scenario, the system's payback period is obtained about 19.5 years with 0.055 M$ net profit. In the second scenario, the purchase cost declines to 0.021 $\$.kWh^{-1}$, and the electricity sale price is assumed to be 0.318 $\$.kWh^{-1}$. Also, the cooling load sale price is increased up to 0.1 $\$.kWh^{-1}$. Considering these prices for purchase and products' sale prices presents 1.41 M$ net profit during 20 years system's lifetime with 7.75 years payback period. Finally, in the third scenario, the electricity purchase cost for charging time is assumed to be about 0.021 $\$.kWh^{-1}$, and the electricity and cooling load's sale prices are considered to be 0.38 $\$.kWh^{-1}$ and 0.12 $\$.kWh^{-1}$, respectively. This

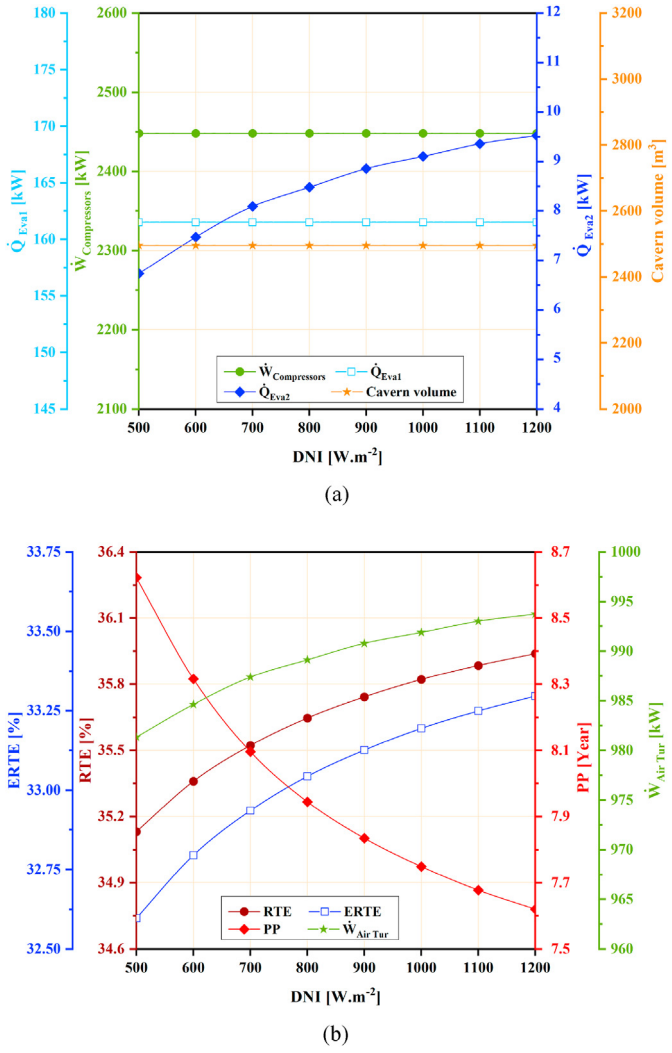


Fig. 15. The effect of DNI variation on the system's performance.

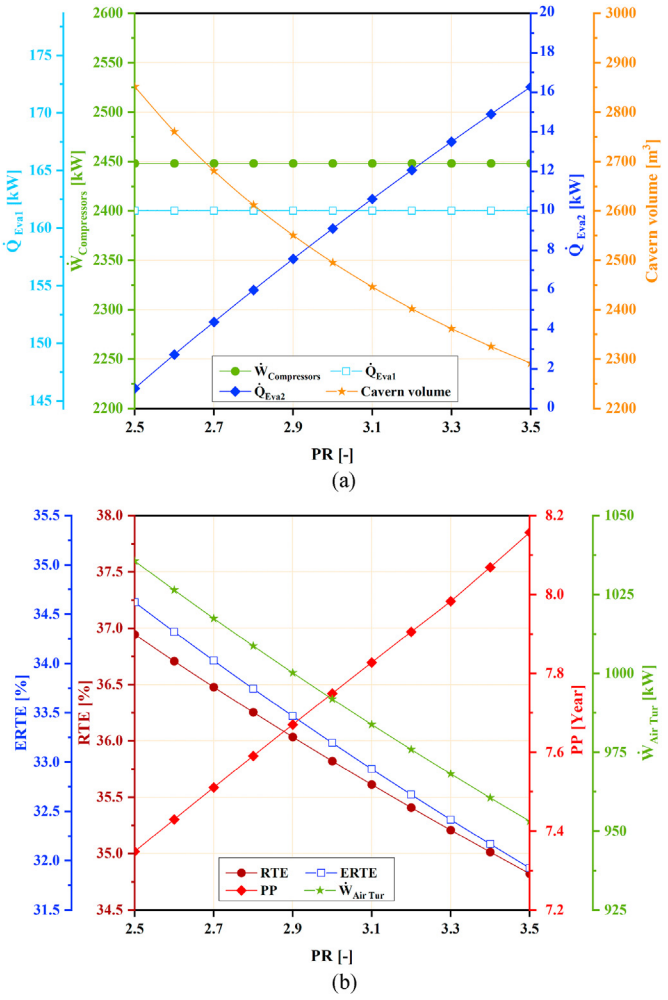


Fig. 16. The effect of charging to discharging pressure ratio on the system's performance.

scenario presents 6 years payback period and a 2.27 M$ net profit. Comparing the second and third scenario's results reveals that increasing the product sale prices by about 20% enhances the payback period and net profit by about 22.58% and 61%, respectively.

5. Optimization

In this section, triple-objective optimization based on the MOPSO method is applied to the proposed system to optimize the payback period, round trip efficiency, and exergetic round trip efficiency. Hence, the vapor generator's temperature, ERC 1's zeotropic mixture's mass fraction, cavern inlet pressure, and charging to discharging pressure ratio (PR) are selected as the optimization decision-maker variables. Also, the variation dominance of the decision-makers variables is presented in Table 9.

Using the MOPSO method for the optimization of the system, the 3-D Pareto-frontier of the optimum states is presented in Fig. 18. The Pareto frontier shows a trade-off between the RTE, ERTE, and payback period. These parameters conflict with each other. To select the best optimum state, a TOPSIS code is applied to the Pareto-frontier. The TOPSIS code selects the 35.87% RTE, 33.23% ERTE, and 7.72 years payback period as the best optimum state, which is shown in Fig. 18. Furthermore, the performance criteria's comparison between the base and optimum state is represented in Table 10. According to Table 10 information, the air turbine power production increases to 997.28 kW, and the ERC1 cooling capacity enhances to 165.70 kW. The ERC 2's cooling capacity reduces to 8.3 kW, but its COP improves to 0.16. Also, the cavern volume declines to 24445.86 m^3.

6. Conclusion

The current study proposed a novel combination of a solar-based A-CAES system in which two ERC subsystems with the zeotropic mixtures were employed to recover the proposed system waste heat and produce a cooling load. The thermodynamic and thermo-economic analyses were performed to evaluate the proposed system performance. Also, eight different zeotropic mixtures were candidates for ERC 1's working fluid. The ERC 1's working fluid selection was conducted through a parametric study to compare

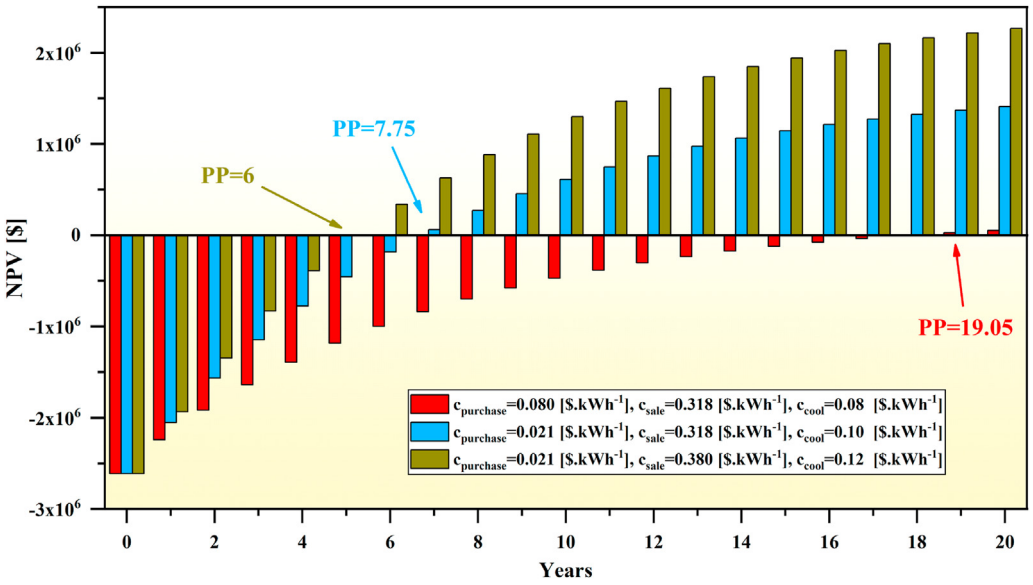


Fig. 17. The net present value diagram of the proposed system for three different purchase and product sale prices.

Table 9
The decision-maker variables' variation range.

Parameter	Variation range
ERC 1's zeotropic mixture's mass fraction	$0 \leq m_{f,ERC\ 1} \leq 1$
Vapor generator evaporation temperature [K]	$350 \leq T_{Gen\ 1} \leq 370$
Vapor generator evaporation temperature [K]	$335 \leq T_{Gen\ 2} \leq 345$
Cavern inlet pressure [kPa]	$3500 \leq P_{cavern} \leq 5500$
Charging to discharging pressure ratio	$2.5 < PR < 3.5$

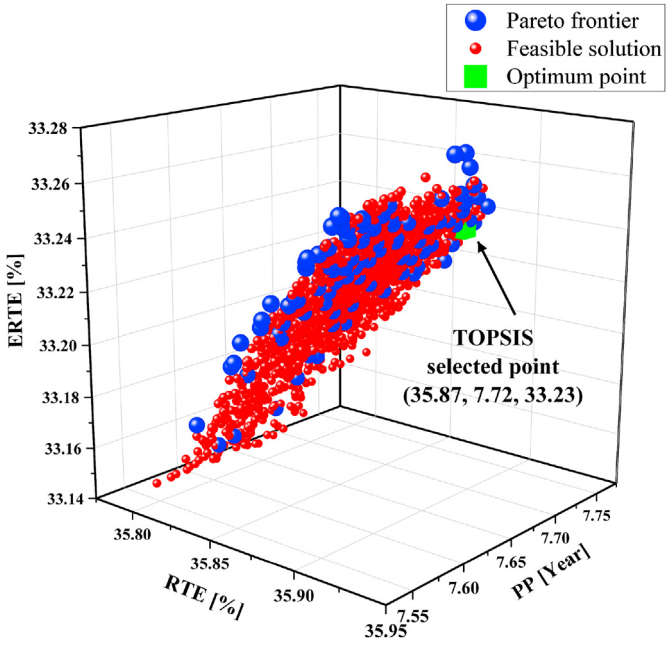


Fig. 18. The 3-D Pareto-frontier and optimization result.

the candidates' performance versus their mass fraction variation. Furthermore, another parametric study was conducted to study the effect of some main parameters, including inlet pressure of air

Table 10
The comparison of the proposed system's performance criteria at the base and optimum state.

Parameter	Base	Optimum
$\dot{W}_{Comp1}$ [kW]	754.79	760.00
$\dot{W}_{Comp2}$ [kW]	832.66	838.96
$\dot{W}_{Comp3}$ [kW]	860.69	867.55
$\dot{W}_{Air\ Tur}$ [kW]	991.89	997.28
$\dot{Q}_{Eva1}$ [kW]	161.5	165.70
COP_{ERC1} [−]	0.16	0.16
$\dot{Q}_{Eva2}$ [kW]	9.2	8.30
COP_{ERC2} [−]	0.11	0.16
$\dot{Q}_{Cooler1}$ [kW]	606.81	611.01
$\dot{Q}_{Cooler2}$ [kW]	795.75	801.77
$\dot{Q}_{Cooler3}$ [kW]	873.45	880.72
$\dot{Q}_{LTES}$ [kW]	133.92	133.98
Cavern volume [m^3]	2495.5	2445.86
$\dot{E}_D^{Charging,Total}$ [kW]	1290.47	1297.48
$\dot{E}_D^{Discharging,Total}$ [kW]	517.99	518.76
PP [Years]	7.75	7.72
RTE [%]	35.82	35.87
ERTE [%]	33.19	33.23
$\eta_{Energy,\ Coll}$ [%]	73.31	73.31
$\eta_{Exergy,Coll}$ [%]	31.64	31.64

cavern ($P_{in\ cavern}$), direct normal irradiation (DNI), and the charging to discharging pressure ratio (PR) variation on the proposed system performance criteria, including cooling load ($\dot{Q}_{Eva1}$ and $\dot{Q}_{Eva2}$), the input power of compressors ($\dot{W}_{Compressors}$), cavern volume, RTE, ERTE, payback period, and power generation capacity ($\dot{W}_{Air\ Tur}$). Then, the NPV was estimated for three different purchase and product sale prices. Finally, a triple-objective optimization was applied to optimize the system's performance. All conducted studies led to the following conclusions:

- The Pentane/Butane with 0.5–0.5 mass fraction presents the highest performance compared to the other considered zeotropic mixtures.

- The proposed system provides 991.89 kW power for peak-shaving of the grid's demand with 35.82% RTE and 33.19% ERTE.
- The DNI has the highest impact on the system's payback period, and the charging to discharging pressure ratio has the greatest influence on the RTE and ERTE.
- Increasing the product's sale prices by about 20% enhances the system's NPV by about 61% and reduces the payback period by about 22.58%.
- Applying the triple-objective optimization enhances the payback period, RTE and ERTE by about 0.38%, 0.05%, and 0.04%, respectively.

Credit author statement

Hao Bai. Methodology, Software, Validation, Writing – review & editing, Language review. ShiHao Luo. Formal analysis, Methodology, Investigation, Writing – review & editing, Language review. Xijie Zhao. Conceptualization, Investigation, Methodology, Supervision, Resources, Data curation. Yang Gao. Writing – review & editing, Language review, Methodology, Investigation, Data curation. Gen Zhao. Writing – review & editing, Language review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Sohani A, Sayyaadi H, Zeraatpisheh M. Optimization strategy by a general approach to enhance improving potential of dew-point evaporative coolers. Energy Convers Manag 2019;188:177–213.
[2] Alirahmi SM, Rostami M, Farajollahi AH. Multi-criteria design optimization and thermodynamic analysis of a novel multi-generation energy system for hydrogen, cooling, heating, power, and freshwater. Int J Hydrogen Energy 2020.
[3] Shakibi H, Afzal S, Shokri A, Sobhani B. Utilization of a phase change material with metal foam for the performance improvement of the photovoltaic cells. J Energy Storage 2022;51:104466.
[4] Budt M, Wolf D, Span R, Yan J. A review on compressed air energy storage: basic principles, past milestones and recent developments. Appl Energy 2016;170:250–68.
[5] Ren J, Ren X. Sustainability ranking of energy storage technologies under uncertainties. J Clean Prod 2018;170:1387–98.
[6] Arabkoohsar A, Rahrabi HR, Alsagri AS, Alrobaian AA. Impact of Off-design operation on the effectiveness of a low-temperature compressed air energy storage system. Energy 2020;197:117176.
[7] Chen S, Arabkoohsar A, Zhu T, Nielsen MP. Development of a micro-compressed air energy storage system model based on experiments. Energy 2020;197:117152.
[8] Cavallo A. Controllable and affordable utility-scale electricity from intermittent wind resources and compressed air energy storage (CAES). Energy 2007;32:120–7.
[9] Kim YM, Favrat D. Energy and exergy analysis of a micro-compressed air energy storage and air cycle heating and cooling system. Energy 2010;35:213–20.
[10] Grazzini G, Milazzo A. Thermodynamic analysis of CAES/TES systems for renewable energy plants. Renew Energy 2008;33:1998–2006.
[11] Luo X, Wang J, Dooner M, Clarke J, Krupke C. Overview of current development in compressed air energy storage technology. Energy Proc 2014;62:603–11.
[12] Jakiel C, Zunft S, Nowi A. Adiabatic compressed air energy storage plants for efficient peak load power supply from wind energy: the European project AA-CAES. Int J Energy Technol Pol 2007;5:296–306.
[13] Kim Y-M, Lee J-H, Kim S-J, Favrat D. Potential and evolution of compressed air energy storage: energy and exergy analyses. Entropy 2012;14:1501–21.
[14] Odukomaiya A, Abu-Heiba A, Gluesenkamp KR, Abdelaziz O, Jackson RK, Daniel C, et al. Thermal analysis of near-isothermal compressed gas energy storage system. Appl Energy 2016;179:948–60.
[15] Arabkoohsar A, Nami H. Thermodynamic and economic analyses of a hybrid waste-driven CHP–ORC plant with exhaust heat recovery. Energy Convers Manag 2019;187:512–22.
[16] Alsagri AS, Arabkoohsar A, Rahbari HR, Alrobaian AA. Partial load operation analysis of trigeneration subcooled compressed air energy storage system. J Clean Prod 2019;238:117948.
[17] Alsagri AS, Arabkoohsar A, Alrobaian AA. Combination of subcooled compressed air energy storage system with an Organic Rankine Cycle for better electricity efficiency, a thermodynamic analysis. J Clean Prod 2019;239:118119.
[18] Safder U, Ifaei P, Yoo C. Multi-objective optimization and flexibility analysis of a cogeneration system using thermorisk and thermoeconomic analyses. Energy Convers Manag 2018;166:602–36.
[19] Hemmati M, Mohammadi-Ivatloo B, Abapour M, Shafiee M. Thermodynamic modeling of compressed air energy storage for energy and reserve markets. Appl Therm Eng 2021;193:116948.
[20] Razmi AR, Afshar HH, Pourahmadiyan A, Torabi M. Investigation of a combined heat and power (CHP) system based on biomass and compressed air energy storage (CAES). Sustain Energy Technol Assessments 2021;46:101253.
[21] Mohammadi A, Ahmadi MH, Bidi M, Joda F, Valero A, Uson S. Exergy analysis of a Combined Cooling, Heating and Power system integrated with wind turbine and compressed air energy storage system. Energy Convers Manag 2017;131:69–78.
[22] Razmi A, Soltani M, Torabi M. Investigation of an efficient and environmentally-friendly CCHP system based on CAES, ORC and compression-absorption refrigeration cycle: energy and exergy analysis. Energy Convers Manag 2019;195:1199–211.
[23] Wang X, Yang C, Huang M, Ma X. Off-design performances of gas turbine-based CCHP combined with solar and compressed air energy storage with organic Rankine cycle. Energy Convers Manag 2018;156:626–38.
[24] Sadeghi S, Ahmadi P. Thermo-economic optimization of a high-performance CCHP system integrated with compressed air energy storage (CAES) and carbon dioxide ejector cooling system. Sustain Energy Technol Assessments 2021;45:101112.
[25] Alirahmi SM, Razmi AR, Arabkoohsar A. Comprehensive assessment and multi-objective optimization of a green concept based on a combination of hydrogen and compressed air energy storage (CAES) systems. Renew Sustain Energy Rev 2021;142:110850.
[26] Javidmehr M, Joda F, Mohammadi A. Thermodynamic and economic analyses and optimization of a multi-generation system composed by a compressed air storage, solar dish collector, micro gas turbine, organic Rankine cycle, and desalination system. Energy Convers Manag 2018;168:467–81.
[27] Zhang X, Zeng R, Deng Q, Gu X, Liu H, He Y, et al. Energy, exergy and economic analysis of biomass and geothermal energy based CCHP system integrated with compressed air energy storage (CAES). Energy Convers Manag 2019;199:111953.
[28] Meng H, Wang M, Aneke M, Luo X, Olumayegun O, Liu X. Technical performance analysis and economic evaluation of a compressed air energy storage system integrated with an organic Rankine cycle. Fuel 2018;211:318–30.
[29] Razmi A, Soltani M, Tayefeh M, Torabi M, Dusseault MB. Thermodynamic analysis of compressed air energy storage (CAES) hybridized with a multi-effect desalination (MED) system. Energy Convers Manag 2019;199:112047.
[30] Sadreddini A, Fani M, Aghdam MA, Mohammadi A. Exergy analysis and optimization of a CCHP system composed of compressed air energy storage system and ORC cycle. Energy Convers Manag 2018;157:111–22.
[31] Razmi AR, Soltani M, Ardehali A, Gharali K, Dusseault MB, Nathwani J. Design, thermodynamic, and wind assessments of a compressed air energy storage (CAES) integrated with two adjacent wind farms: a case study at Abhar and Kahak sites, Iran. Energy 2021;221:119902.
[32] Mousavi SB, Adib M, Soltani M, Razmi AR, Nathwani J. Transient thermodynamic modeling and economic analysis of an adiabatic compressed air energy storage (A-CAES) based on cascade packed bed thermal energy storage with encapsulated phase change materials. Energy Convers Manag 2021;243:114379.
[33] Alirahmi SM, Mousavi SB, Razmi AR, Ahmadi P. A comprehensive techno-economic analysis and multi-criteria optimization of a compressed air energy storage (CAES) hybridized with solar and desalination units. Energy Convers Manag 2021;236:114053.
[34] Li W, Feng X, Yu LJ, Xu J. Effects of evaporating temperature and internal heat exchanger on organic Rankine cycle. Appl Therm Eng 2011;31:4014–23.
[35] Wang XD, Zhao L. Analysis of zeotropic mixtures used in low-temperature solar Rankine cycles for power generation. Sol Energy 2009;83:605–13.
[36] Zhao L, Bao J. Thermodynamic analysis of organic Rankine cycle using zeotropic mixtures. Appl Energy 2014;130:748–56.
[37] Klein SA, Alvarado F. Engineering equation solver, version 9.083. Middleton, WI: F-Chart Software; 2012.
[38] Lemmon EW, Bell IH, Huber ML, McLinden MO. NIST standard reference database 23: reference fluid thermodynamic and transport properties-REFPROP, version 10.0. Gaithersbg: National Institute of Standards and Technology. Stand Ref Data Program; 2018.
[39] Lemmon EW, Huber ML, McLinden MO. NIST reference fluid thermodynamic and transport properties–REFPROP. NIST Stand Ref Database 2002;23. v7.
[40] Ahmadi S, Ghaebi H, Shokri A. A comprehensive thermodynamic analysis of a novel CHP system based on SOFC and APC cycles. Energy 2019;186:115899.
[41] Hamrang F, Shokri A, Seyed Mahmoudi SM, Ehghaghi B, Rosen MA. Performance analysis of a new electricity and freshwater production system based on an integrated gasification combined cycle and multi-effect desalination. Sustain Times 2020;12.
[42] Rostamzadeh H, Ebadollahi M, Ghaebi H, Shokri A. Comparative study of two

novel micro-CCHP systems based on organic Rankine cycle and Kalina cycle. Energy Convers Manag 2019;183:210–29.
[43] Nedaei N, Hamrang F, Farshi LG. Design and 3E analysis of a hybrid power plant integrated with a single-effect absorption chiller driven by a heliostat field: a case study for Doha. Qatar Energy 2021:122415.
[44] Yao Z, Yuxing Z, Yaqian K, Sobhani B. Research on an integrated power and freshwater generation system from natural gas energy and geothermal sources. Desalination 2022;525:115494.
[45] Ziapour BM, Saadat M, Palideh V, Afzal S. Power generation enhancement in a salinity-gradient solar pond power plant using thermoelectric generator. Energy Convers Manag 2017;136:283–93.
[46] Nedaei N, Azizi S, Farshi LG. Performance assessment and multi-objective optimization of a multi-generation system based on solar tower power, A case study in Dubai. UAE Process Saf Environ Prot 2022.
[47] Parikhani T, Jannatkhah J, Shokri A, Ghaebi H. Thermodynamic analysis and optimization of a novel power generation system based on modified Kalina and GT-MHR cycles. Energy Convers Manag 2019;196:418–29.
[48] Puttur U, Ahmadi M, Ahmadi B, Bigham S. A novel lung-inspired 3D-printed desiccant-coated heat exchanger for high-performance humidity management in buildings. Energy Convers Manag 2022;252:115074.
[49] Razmi A, Soltani M, Aghanajafi C, Torabi M. Thermodynamic and economic investigation of a novel integration of the absorption-recompression refrigeration system with compressed air energy storage (CAES). Energy Convers Manag 2019;187:262–73.
[50] Soltani M, Nabat MH, Razmi AR, Dusseault MB, Nathwani J. A comparative study between ORC and Kalina based waste heat recovery cycles applied to a green compressed air energy storage (CAES) system. Energy Convers Manag 2020;222:113203.
[51] Nabat MH, Sharifi S, Razmi AR. Thermodynamic and economic analyses of a novel liquid air energy storage (LAES) coupled with thermoelectric generator and Kalina cycle. J Energy Storage 2022;45:103711.
[52] Moghimi M, Emadi M, Ahmadi P, Moghadasi H. 4E analysis and multi-objective optimization of a CCHP cycle based on gas turbine and ejector refrigeration. Appl Therm Eng 2018;141:516–30.
[53] Zhou Y, Li S, Sun L, Zhao S, Talesh SSA. Optimization and thermodynamic performance analysis of a power generation system based on geothermal flash and dual-pressure evaporation organic Rankine cycles using zeotropic mixtures. Energy 2020;194:116785.
[54] Mahdavi N, Ghaebi H, Minaei A. Proposal and multi-aspect assessment of a novel solar-based trigeneration system; investigation of zeotropic mixture's utilization. Appl Therm Eng 2022:118110.
[55] Ding P, Zhang K, Yuan Z, Wang Z, Li D, Chen T, et al. Multi-objective optimization and exergoeconomic analysis of geothermal-based electricity and cooling system using zeotropic mixtures as the working fluid. J Clean Prod 2021:126237.
[56] Cengel YA, Boles MA, Kanoglu M. Thermodynamics: an engineering approach, vol. 5. New York: McGraw-hill; 2011.
[57] Huang Y, Jiang P, Zhu Y. Experimental and modeling studies of thermally-driven subcritical and transcritical ejector refrigeration systems. Energy Convers Manag 2020;224:113361.
[58] Eshaghi S, Hamrang F. An innovative techno-economic analysis for the selection of an integrated ejector system in the flare gas recovery of a refinery plant. Energy 2021:120594.
[59] Al-Sulaiman FA, Hamdullahpur F, Dincer I. Performance assessment of a novel system using parabolic trough solar collectors for combined cooling, heating, and power production. Renew Energy 2012;48:161–72.
[60] Al-Sulaiman FA, Dincer I, Hamdullahpur F. Exergy modeling of a new solar driven trigeneration system. Sol Energy 2011;85:2228–43.
[61] Kalogirou SA. Solar energy engineering: processes and systems. Academic Press; 2013.
[62] Çengel YA, Boles MA. Thermodynamics: an engineering approach. eight edition 2015.
[63] Bejan A, Tsatsaronis G, Moran MJ. Thermal design and optimization. John Wiley & Sons; 1995.
[64] Abam FI, Briggs TA, Diemuodeke OE, Ekwe EB, Ujoatuonu KN, Isaac J, et al. Thermodynamic and economic analysis of a Kalina system with integrated lithium-bromide-absorption cycle for power and cooling production. Energy Rep 2020;6:1992–2005.
[65] Emadi MA, Chitgar N, Oyewunmi OA, Markides CN. Working-fluid selection and thermoeconomic optimisation of a combined cycle cogeneration dual-loop organic Rankine cycle (ORC) system for solid oxide fuel cell (SOFC) waste-heat recovery. Appl Energy 2020;261:114384.
[66] Razmi AR, Janbaz M. Exergoeconomic assessment with reliability consideration of a green cogeneration system based on compressed air energy storage (CAES). Energy Convers Manag 2020;204:112320.
[67] Habibi H, Zoghi M, Chitsaz A, Shamsaiee M. Thermo-economic performance evaluation and multi-objective optimization of a screw expander-based cascade Rankine cycle integrated with parabolic trough solar collector. Appl Therm Eng 2020;180:115827.
[68] Wu S, Zhou C, Doroodchi E, Moghtaderi B. Techno-economic analysis of an integrated liquid air and thermochemical energy storage system. Energy Convers Manag 2020;205:112341.
[69] Liu F, Yang C, Li B, Silang Y, Zhu Y, Farkoush SG. Thermodynamic and economic sensitivity analyses of a geothermal-based trigeneration system; performance enhancement through determining the best zeotropic working fluid. Energy 2022;246:123310.